

The Industrial Organization of Belief: Market Size, Church Expansion, and Religious Change

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Abstract

We study how migration reallocates market shares across organizations by expanding local market scale. When institutions differ in how effectively capacity expansion generates new demand, population inflows can redistribute market shares rather than scale providers proportionally. We examine this mechanism in Brazil, where internal migration between 1980 and 2010 coincided with a major religious transition. Using a shift-share design based on agricultural price shocks, we document a *Composition Paradox*: although migrants originated from Catholic strongholds, their arrival reduced Catholic and increased Evangelical market shares among natives, ruling out cultural transmission. Instead, migration induced an asymmetric organizational response: Evangelical denominations rapidly expanded temple infrastructure and converted this expansion into affiliation, while the Catholic Church did not. A structural model shows that the elasticity of affiliation with respect to temple expansion is higher for Evangelicals, reflecting a more effective conversion and proselytism technology that raises the returns to expansion and translates infrastructure growth into affiliation. More broadly, the results illustrate a general principle: rapid expansions in market size favor organizations whose technologies more effectively translate capacity into market share.

Keywords: Migration; Religion; Market-Size Shocks; Demand Elasticity.

JEL Codes: J61; Z12; O15; D02; L31.

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1 Introduction

Population movements are among the most consequential shocks affecting modern societies, reconfiguring markets through mechanisms that go beyond simple composition effects. While recent scholarship shows that migrants can reshape economic and social outcomes by importing specific cultural traits (Bazzi et al., 2023), we argue that migration also operates through a distinct economic channel: by expanding market size. Even absent cultural differences, migrant inflows increase the scale of local markets, altering entry and expansion incentives for organizations. These changes can reshape equilibrium market structure and reallocate market share across organizations, even when underlying preferences remain unchanged.

We document this in Brazil, where internal migration coincided with a historic religious transition. Between 1980 and 2010, the Catholic share of the population fell from 90% to 65%, accompanied by profound shifts in social norms, civic life, and politics (Corbi and Sanches, 2024; Costa et al., 2019; Cavalcanti et al., 2022), mirroring trends across the Global South (Pew, 2006). This context offers a rare opportunity to study how demographic shocks interact with organizational frictions to generate durable institutional change. Crucially, it allows us to rule out cultural transmission through what we term the *Composition Paradox*: although migrants originated from Catholic strongholds, their arrival triggered large increases in Evangelical adherence and political support for Evangelical candidates among native residents.

This counter-intuitive pattern implies that migration did not operate by importing beliefs. Instead, our analysis suggests that population inflows relaxed scale constraints and reallocated market share toward denominations whose infrastructure expansions generated larger demand responses.¹ In our framework, this higher demand elasticity with respect to infrastructure simultaneously implies higher returns to investment and greater affiliation gains, explaining both the rapid expansion of Evangelical capacity and its effectiveness in converting scale into market share. This perspective places industrial organization—rather than preferences or beliefs—at the center of the analysis.

We distinguish between two economic consequences of population inflows. First, the

¹See also recent work emphasizing market size as a first-order determinant of organizational and technological outcomes (Goldberg and Reed, 2023).

Income Channel: migration acts as a labor supply shock that can depress local wages. If religious adherence provides social insurance, negative income shocks increase the demand for affiliation, disproportionately benefiting denominations that bundle spiritual services with dense support networks and mutual aid. Second, the *Religious Infrastructure Channel:* migration expands the scale of the local religious market, increasing the incentives to invest in local physical presence. Because denominations differ in how strongly new temples translate into affiliation, population inflows generate an asymmetric expansion of religious infrastructure, reallocating market share toward denominations that convert population growth into local temples and adherents more effectively.

To evaluate these competing mechanisms, we treat internal migration as an exogenous market-size shock and study its effect on the religious choices of native residents. Our empirical strategy uses a shift-share instrumental variable (SSIV) design that exploits international commodity-price shocks at rural origins as exogenous “push factors.” Declining agricultural prices depress incomes in origin municipalities and induce out-migration. Predicted migrant outflows are allocated across destinations using historical migration networks. Following Borusyak et al. (2022), we implement the origin-shock transformation of the SSIV estimator, ensuring valid inference in the presence of correlated exposure shares (Adao et al., 2019). Because the shocks arise in rural origins, our instrument is orthogonal to religious amenities or proselytizing activity in destination areas, minimizing concerns about cultural-transmission confounds.

The empirical study proceeds in three steps. First, we document that migration reduced Catholic affiliation and increased Evangelical affiliation among natives. Our SSIV estimates show that between 1980 and 2010 migration explains roughly 3.7 percentage points—about 23%—of the national rise in Evangelicalism. The shift in religious market share translated into political behavior: municipalities more exposed to migration exhibited significantly higher vote shares for Evangelical candidates, consistent with deep, persistent behavioral changes rather than nominal reclassification.

Second, we trace these effects to two distinct economic channels generated by the market-size shock. On the income channel, we find that migration reduced local earnings by approximately 5% on average in each decennial period covered by our data, potentially increasing

demand for religious insurance, which Evangelical denominations are, according to the literature, better positioned to provide (Cavalcanti et al., 2022).² On the religious infrastructure channel, migration expanded the scale of the local religious market and triggered an asymmetric entry of temples. Evangelical denominations consistently and substantially increased the number of local temples in response to population inflows, while the Catholic Church did not. Absent migration, Evangelical temple density would be approximately 27% lower.

To assess the robustness of these results and address concerns about the validity of the instrument, we conduct a series of additional tests across all exercises. First, we include lagged values of the dependent variable to account for potential pre-trends in religious composition across municipalities. Second, we control for lagged predicted migration to mitigate concerns that our estimates may reflect persistent migration flows rather than contemporaneous shocks. Third, we also control for differential trends in economic and demographic variables—such as population size, income per capita, education, racial composition, and age structure—as well as state fixed effects, which absorb broader regional religious trends. Fourth, we account for the possibility that price shocks in origin municipalities could influence destination outcomes through other channels—such as trade or capital reallocation—by including in our models a measure of exposure of destinations to shocks in other municipalities, aggregating shocks in other municipalities weighted by the inverse of the distance between destination and other municipalities. Finally, we include local food price shocks at destination as an additional control to ensure that our estimates are not confounded by local economic fluctuations that may be correlated with broader commodity cycles. While no observational strategy can eliminate all concerns about identification, the robustness of our results across multiple specifications lends support to a causal interpretation of our results.

Finally, to shed light on the sources of the asymmetric expansion of Evangelical temples and to quantify the relative importance of the income and infrastructure channels, we develop a structural model of religious demand in the spirit of Berry et al. (2004); see also Corbi et al. (2022) for an application to the U.S. religious market. Individuals choose between Catholicism, Evangelicalism, and no religion based on their income and the local religious

²More generally, a large literature links Evangelical growth to economic distress (Chesnut, 1997; Costa et al., 2019). Internal migration has also been shown to reduce earnings in Brazil (Corbi et al., 2025; Imbert and Ulyssea, 2022).

infrastructure available to them. Estimates show that Evangelical demand is highly elastic with respect to temple availability and negatively related to income, while Catholic demand is comparatively insensitive to both margins. This higher elasticity implies both stronger incentives to expand temple infrastructure and larger affiliation gains following expansion, jointly rationalizing the observed temple growth and market-share reallocation toward Evangelicals. Independent difference-in-differences estimates exploiting variation in temple entry corroborate these patterns.³ Counterfactual simulations show that the infrastructure channel accounts for the vast majority of the observed religious shift, overwhelmingly dominating the income channel: reproducing the transition through income shocks alone would require implausible wage declines, whereas the observed expansion of Evangelical infrastructure fully rationalizes the transition.⁴ The limited role of the income channel thus reflects not only the modest magnitude of migration-induced income shocks, but also the low sensitivity of Evangelical affiliation to income.

This paper makes two contributions to the literature. First, we advance the still nascent literature on how migration transforms religious markets. Bazzi et al. (2023) emphasize the cultural transmission of religious and political preferences by migrants, while Hersey (2024) shows how Evangelicals gained market share by integrating and supporting migrants in Ghana. Our approach centers on a different mechanism: internal migration expands local market size, inducing asymmetric expansion of religious infrastructure that reshapes affiliation patterns among native populations. This links our findings to broader work on how structural factors—such as regulation, public policy, and resource reallocation—affect the evolution of religious markets (Barro and McCleary, 2005; Gruber and Hungerman, 2008; Hungerman, 2005; Cantoni et al., 2018; Bazzi et al., 2020).

Second, we contribute to the debate over the rise of Evangelicalism in Brazil and other regions by clarifying the relative roles of demand- and supply-side mechanisms. While much of the literature stresses poverty, exclusion, and economic distress as demand-side drivers operating through income effects (Mariano, 2011; Chesnut, 1997; Martin, 1990; Costa et al.,

³Using Callaway and Sant’Anna (2021), we find that Evangelical temple openings cause sustained increases in Evangelical affiliation with no pre-trends, while Catholic openings have no significant effects.

⁴Because we do not explicitly model temple supply, this exercise abstracts from potential indirect effects of income on Evangelical affiliation operating through changes in temple entry. Given the weak sensitivity of demand to income, such effects are likely to be small and do not affect our main conclusions.

2019; Cavalcanti et al., 2022), a complementary line of research emphasizes organizational and institutional asymmetries across denominations (Finke and Stark, 1988; Stark and Finke, 2000; Iannaccone et al., 1997; Stark and Iannaccone, 1994; Putnam et al., 2012). Our results provide empirical support for this latter view.

The remainder of the paper is organized as follows. Section 2 describes the institutional context and the data. Section 3 outlines the empirical strategy and presents the baseline effects of migration on religious affiliation and political outcomes. Section 4 examines the income and infrastructure channels. Section 5 develops the structural model and quantifies the importance of each channel. Section 6 concludes.

2 Institutional Background and Data

This section describes the institutional setting and the data used in the analysis. We highlight the features of Brazil’s religious market that make it a suitable environment for studying how population shocks reshape institutional landscapes, and we outline the datasets used to measure religious affiliation, migration flows, church infrastructure, and local economic conditions.

2.1 Data Sources

Our primary data come from the Brazilian Demographic Censuses of 1980, 1991, 2000, and 2010, which provide information on religious affiliation, migration histories, and socioeconomic characteristics at the individual level. We harmonize these data at the municipality level, using the 1980 municipal boundaries as our reference throughout the analysis. Because Brazil’s municipal map changed substantially after 1980 due to subdivision and reclassification, we aggregate all municipalities created after 1980 back to their original 1980 units.⁵

In each Census wave, respondents were asked “What is your religion?” with a set of categories that expanded over time as Brazil’s religious diversity grew. We classify respondents into two main groups: *Catholic* and *Evangelical*. The Evangelical category combines Main-

⁵Using shapefiles for 1980 and 2010, we assign each post-1980 municipality to its 1980 counterpart based on the location of its centroid.

line Protestants and the broader set of Evangelical denominations, including Pentecostal, non-Pentecostal, and non-denominational churches (see Table A1 in the Appendix).⁶ Other religious groups—Spiritist, Afro-Brazilian, Jewish, other non-Christian religions, and those reporting no religion—collectively represent less than 5% of the population and are not the focus of our analysis.

The Censuses also collect information on individuals’ prior residence. Because the reference period differs across waves, we construct a consistent measure of migration based on moves within the previous five years, following Imbert and Ulyssea (2022) and Corbi et al. (2025). In each Census, a migrant is defined as an individual living in a municipality different from the one in which they resided five years earlier. These data allow us to construct (i) migration inflows at the municipality level and (ii) historical migrant “shares” used in the shift-share instrumental variable (SSIV) strategy described in Section 3. In addition, we calculate Census-based measures of local income, education, population size, racial composition, and age structure. These serve as controls in some specifications and as key outcomes when examining mechanisms.

To study institutional supply responses, we use establishment-level administrative data from the Brazilian Internal Revenue Service (Receita Federal). The registry includes all legal entities in the country, including religious organizations classified under the CNAE category for religious entities (BRASIL, 1991-2018). For each establishment, we observe its denomination, municipality, and opening and closure dates. We classify each establishment as Catholic, Evangelical, or “Other,” and aggregate these data into a municipality-year panel capturing the active stock of temples by religious group. These measures allow us to quantify how institutional supply responded to migration-induced market expansion.

To construct the “shifters” of our SSIV, we combine data on international commodity prices with measures of local agricultural production. Following Imbert and Ulyssea (2022), we use monthly world prices for 12 major commodities (banana, cocoa, coffee, cotton, maize, orange, rice, soybeans, sugar, tobacco, wheat, and wood) from the World Bank’s Pink Sheet (1970–2010). We aggregate these price series into municipality-specific income shocks using pre-determined production shares derived from the Agricultural Census and the Muni-

⁶For details across Census waves, see the IBGE documentation at <https://www.ibge.gov.br/>.

pal Agricultural Survey (IBGE). These municipality-level shocks constitute the exogenous “push factors” for out-migration in our SSIV strategy, detailed in Section 3. Crucially, these rural origin municipalities were historically the strongholds of Brazilian Catholicism, with affiliation rates significantly higher than the national average.

2.2 The Brazilian Religious Market

For most of the twentieth century, the Brazilian religious market was characterized by the near-monopoly of the Catholic Church. The subsequent rapid expansion of Evangelicalism—a category that includes Mainline Protestants and Pentecostals—reflects the diffusion of a distinct mode of religious production centered on intensive proselytism and conversion. Unlike the Catholic Church, whose model emphasizes sacramental provision and territorial parishes, Evangelical denominations have invested heavily in practices that actively recruit new adherents, including frequent services, personalized outreach, lay participation, and the extensive use of mass media and religious broadcasting, alongside direct engagement with potential converts (Sola, 2024; Komatsu, 2019).

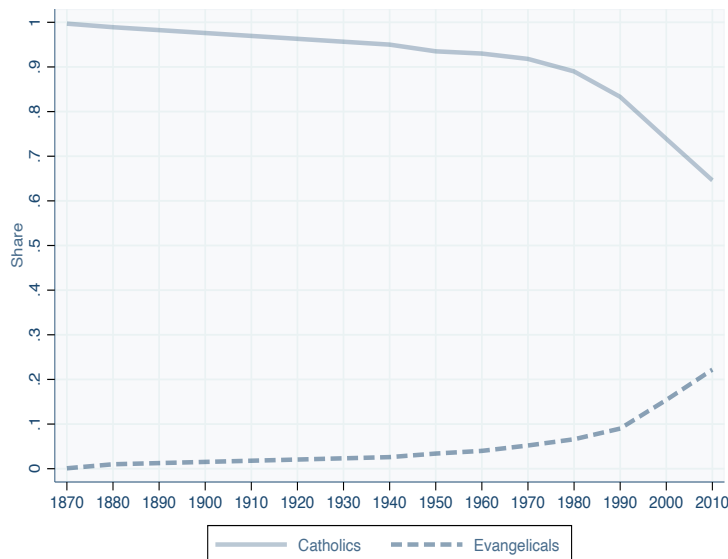
This emphasis on proselytism has allowed Evangelical churches to translate local presence into active affiliation with high effectiveness. Local units are designed not merely to serve existing members but to actively recruit new ones, making affiliation highly responsive to the expansion of religious infrastructure. As a result, increases in Evangelical presence are closely followed by increases in Evangelical affiliation.

Figure 1 illustrates the resulting reallocation of market shares observed during the last decades. Between 1980 and 2010, the Catholic share of the population fell from approximately 90% to 65%, while the Evangelical share rose from 6% to 22%. This shift represents not merely a change in private belief but a massive institutional reorganization with significant political consequences, as Evangelical groups have increasingly mobilized into a cohesive voting bloc (Freston, 2001; Costa et al., 2019). We provide a detailed breakdown of denominations and their historical evolution in Appendix A.

These institutional features motivate our empirical design, in which we treat internal migration as an exogenous market-size shock and examine how it affects religious choices of native residents. We now outline the empirical strategy used to isolate the causal effect of

migration.

Figure 1: Shares of Evangelicals and Catholics in Brazil (1870–2010)



Notes: This figure shows the shares of Catholics and Evangelicals (including Mainline, Pentecostal, and non-Pentecostal denominations). Data from Alves et al. (2014), based on the Brazilian Censuses of 1870, 1880, 1940, 1950–1980, 1991, 2000, 2010 (IBGE). Data for 1890–1930 interpolated linearly.

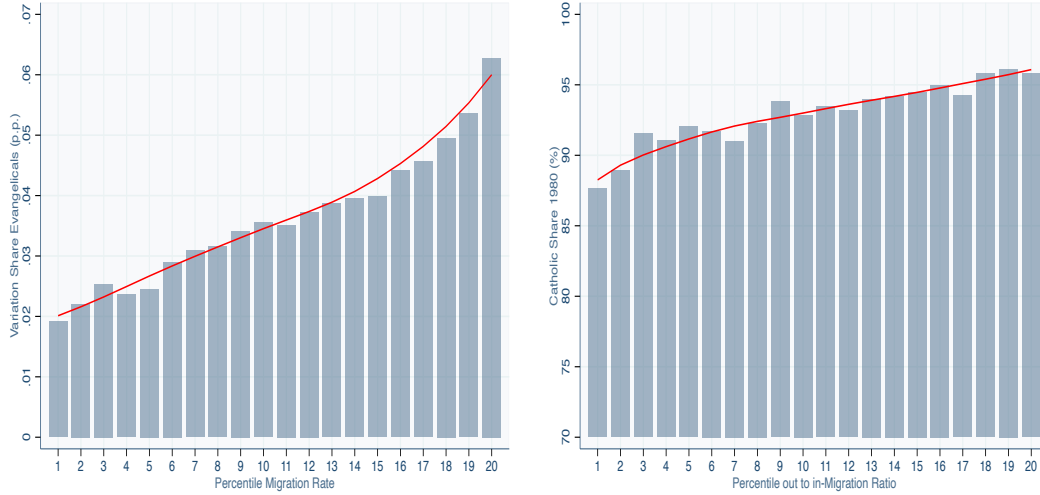
3 Internal Migration and the Rise of Evangelicals

This section develops the empirical framework used to estimate the causal effect of migration-induced population shocks on religious affiliation. Because migration may affect local labor markets, neighborhood composition, and institutional supply, we focus on how these broader equilibrium effects influence the religious choices of native individuals who did not move. We begin with descriptive evidence and then introduce our identification strategy based on shift-share instrumental variables.

Figure 2 presents two key correlations. Panel (a) shows a strong positive association between in-migration rates and the rise in the Evangelical share among natives. Panel (b) shows that cities with higher out-migration relative to in-migration rates between 1980 and 2010 had higher initial Catholic shares—evidence consistent with what we termed the *Composition Paradox*, namely that migrant-sending regions were more predominantly Catholic than the regions receiving them.

Although purely descriptive, these patterns align with the mechanism outlined in the Introduction: the rise of Evangelicalism in Brazil does not appear to stem from direct cultural transmission. Rather, it seems to emerge from a more complex economic dynamic—one that we examine in detail throughout the paper.

Figure 2: Migration Rates, Initial Conditions and Growth of Evangelicals Among Native Residents



(a) in-Migration and Evangelical Growth (b) out-Migration and 1980 Catholic Share

In Panel (a) (Panel (b)) each bar represents the average change in the Evangelical share among native residents (the average share of Catholics among native residents in 1980) for municipalities in a given percentile of in-migration rates (in-migration to out-migration ratio). The line shows the LOWESS-smoothed series. Native residents exclude individuals classified as migrants, following IBGE definitions. Data from the Brazilian Censuses, 1980–2010.

3.1 Empirical Model

For each destination municipality d , we estimate regressions of the form:

$$\Delta s_d = \alpha + \beta m_d + \gamma X_d + \epsilon_d, \quad (1)$$

where Δs_d is the change in the share of native residents affiliated with a given religious group between two Census waves, m_d is the accumulated migration inflow over the corresponding period, and X_d includes predetermined demographic controls. We estimate equation (1) separately for the three inter-Census periods (1980–1991, 1991–2000, 2000–2010) to allow

the relationship between migration and religious change to vary over time.

The coefficient β captures the reduced-form effect of migration on religious composition. It aggregates multiple channels— e.g. changes in labor market conditions and institutional supply—that we analyze in subsequent sections. Ordinary least squares estimates of β may be biased because migrants select locations based on economic prospects, and economic conditions are correlated with religious conversion (Costa et al., 2019). To address endogeneity, we construct an instrument for m_d using exogenous shocks to out-migration from rural origins.

3.2 Shift-Share Instrument Construction

Our SSIV follows the “push-factor” design of Bazzi et al. (2023), Boustan (2010), and Derenoncourt (2022). We use international commodity price fluctuations as shocks to agricultural income in origin municipalities and allocate the resulting outflows to destinations using historical settlement patterns.

Origin-level shocks. Drawing on the approach of Imbert and Ulyssea (2022), we construct agricultural income shocks at the origin municipality o :

$$z_{oy}^p = \sum_l \sum_c (\pi_{oc} \times h_o \times \varepsilon_{cly}), \quad (2)$$

where ε_{cly} is the residual from an AR(1) process applied to world prices for commodity c in month l of year y , π_{oc} is the crop-share weight for origin o , and h_o scales shocks by the importance of agriculture in the municipality.

We estimate:

$$m_{oy} = \kappa + \eta z_{o,y-1}^p + \phi_o + \delta_y + \xi_{oy}, \quad (3)$$

where m_{oy} is the out-migration rate from origin o in year y . The predicted number of migrants leaving origin o is:

$$\widehat{M}_{oy} = \widehat{m}_{oy} \times P_o, \quad (4)$$

where P_o is the 1980 population.

Allocation to destinations. Using migration patterns from the 1980 Census, we allocate predicted outflows across destinations:

$$\widehat{M}_{dy} = \sum_{y-5 < t \leq y} \left(\sum_{o=1}^O \nu_{od} \times \widehat{M}_{ot} \right), \quad (5)$$

where ν_{od} is the share of migrants residing in d who originated from o in 1980. We normalize by the 1980 native population in d to obtain the predicted in-migration rate:

$$\widehat{m}_{dy} = \frac{\widehat{M}_{dy}}{P_d}. \quad (6)$$

This SSIV isolates variation arising from exogenous income shocks in rural origins, combined with predetermined historical links, and is plausibly uncorrelated with unobserved determinants of religious change in destinations.

Our SSIV addresses the selection problem by isolating variation in migration driven exclusively by economic “push factors” in rural origins—specifically, negative shocks to agricultural commodity prices (Imbert and Ulyssea, 2022). Unlike migrants who might select destinations based on religious amenities, migrants displaced by global price drops are effectively “accidental” movers. Consequently, the variation we exploit is plausibly orthogonal to the unobserved religious zealotry of the movers. Moreover, as suggested by Figure 2 migrants originated from overwhelmingly Catholic regions, effectively rendering cultural-transmission theories unable to associate the inflow of such migrants with a sizable decline in Catholic affiliation.

Beyond selection, our identification strategy requires that commodity price fluctuations in origin regions be uncorrelated with other unobserved factors influencing religious choices in destination areas (the exclusion restriction). A key concern is that origin shocks might affect the destination economy through channels other than migration, such as trade linkages or general equilibrium price effects. We address these concerns by showing that our results are robust to controlling for trade-weighted shocks, destination-specific price indices, and pre-trends in the outcome variables.

Finally, for inference, we follow Borusyak et al. (2022) and implement the SSIV estimator

at the level of origin shocks rather than destination units. This transformation yields identical point estimates for β but ensures correct inference by accounting for correlations in exposure shares (Adao et al., 2019).

3.3 Price Shocks and out-Migration

Table 1 displays estimates of the relationship between agricultural price shocks and migration outflows from the origin municipalities, represented by equation (3) under various model specifications. The first two columns present the coefficients from the regression of out-migration from each municipality on our measure of commodity price shocks. In the third column, we extend the analysis by interacting price shocks with dummies for positive and negative shocks. All specifications include time and municipality fixed effects. Columns 2-3 also include baseline covariates (share of population under 18 years old; share of non-white; share of female; share with high school education or higher; share of married individuals; household income per capita, and total population) measured in 1980 and interacted with time dummies. Standard errors in parentheses are clustered at the municipality level.

In specifications 1 and 2, the coefficients are negative and statistically significant at the 1% level, indicating that negative shocks to the prices of agricultural products produced in municipality o lead to an increase in (out)migration from that municipality to other areas. The inclusion of covariates has little effect on the point estimates. The third column further reveals that the effect of price shocks on migration is highly asymmetric, with migration responding much more strongly to negative shocks than to positive ones.⁷ One potential concern is that price shocks for certain commodities—particularly soy—may induce structural changes in local economies (Bustos et al., 2016), thereby confounding their relationship with out-migration flows.⁸ To address this, we re-estimate the specifications, excluding soy from the commodity price basket. As shown in columns 4–6, the results remain very similar.

Figure 3 plots observed migration outflows in each municipality alongside those predicted

⁷In principle, both negative and positive price shocks at origin municipalities may induce migration outflows. While negative shocks can force individuals to seek better economic conditions elsewhere, positive shocks can ease financial constraints for those who were already considering migration, allowing them to pursue opportunities in other regions (Bazzi, 2017).

⁸Soy is by far the most economically significant commodity in our baseline specification.

Table 1: Migration Outflows Induced by Price Shocks

	(1)	(2)	(3)	(4)	(5)	(6)
Price shocks _{y-1}	-0.005*** (0.001)	-0.004*** (0.001)				
Negative shocks _{y-1}			-0.007** (0.004)			
Positive shocks _{y-1}			-0.000 (0.003)			
Price shocks _{y-1} (No soy)				-0.004*** (0.001)	-0.002** (0.001)	
Negative shocks _{y-1} (No soy)						-0.005* (0.003)
Positive shocks _{y-1} (No soy)						0.001 (0.003)
F Statistic	11.8	28.2	28.5	9.2	28.6	29.3
Observations	58,905	58,905	58,905	58,905	58,905	58,905
Municipalities	3,927	3,927	3,927	3,927	3,927	3,927
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Baseline covariates		✓	✓		✓	✓

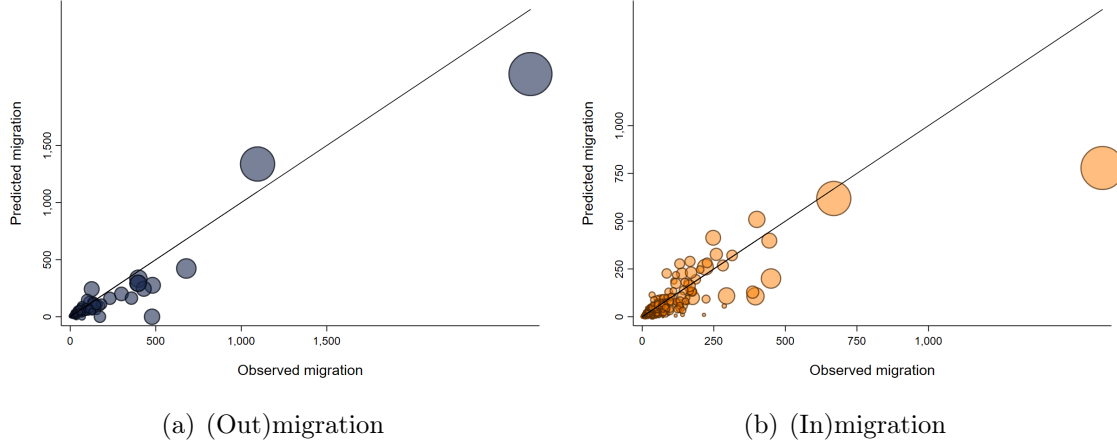
Each observation is an origin municipality by year. The dependent variable is the number of individuals who left the origin area, divided by its total population in 1980. All specifications include time and municipality fixed effects. Columns 2-3 and 5-6 also include baseline covariates (share of population under 18 years old; share of non-white; share of female; share with high school education or higher; share of married individuals; household income per capita, and total population) measured in 1980 and interacted with time dummies. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

by the model in column 2 of Table 1, as well as observed migration inflows and model predictions. Observations are weighted by municipality population in 1980, with the size of each circle proportional to these weights. Overall, the figure indicates that our migration model provides a strong fit for both migration inflows and outflows, suggesting a robust first-stage for our IV estimates.

3.4 Migration Shocks and Religious Composition

We now turn to the relationship between migration rates and changes in the religious composition of native residents within each municipality. Table 2 summarizes these results. Columns report estimates from both OLS and SSIV regressions, with outcomes corresponding to changes in the share of Catholics (columns 1 and 2) and Evangelicals (columns 3 and

Figure 3: **Predicted vs. Observed Migration Flows**



This figure illustrates the relationship between predicted and observed migration flows across Brazilian municipalities from 1985 to 2010. Panel (a) depicts the number of migrants leaving their hometowns, while Panel (b) presents the number of incoming migrants in destination areas. The size of each circle corresponds to the total population of the municipality in 1980. Data source: Brazilian Census.

4). Panel A covers the period 1980–1991, Panel B covers 1991–2000, and Panel C covers 2000–2010. All regressions include the baseline covariates described earlier. OLS estimates come from regressions at the destination-municipality level, while the SSIV regressions are at the origin-municipality level. Standard errors clustered at the respective municipality-level.

The first-stage F-statistics, shown at the bottom of Panels A–C, are very large, varying between 199.1 and 68.4.⁹ These values confirm the strong first-stage effects discussed earlier in this section. Comparisons between OLS and SSIV estimates suggest that the OLS estimates are systematically biased toward zero. One plausible explanation for this bias is that migrants tend to select regions with stronger economic performance, where the Catholic Church performs particularly well, while Evangelical churches perform relatively worse. This mechanism, discussed in Costa et al. (2019), aligns with the observed patterns and highlights the importance of addressing endogeneity in this analysis.

Our analysis of the SSIV estimates in Table 2 reveals two key findings. First, higher migration rates consistently lead to a decline in the share of Catholics across all periods analyzed. Second, this reduction in Catholic affiliation is accompanied by a statistically significant increase in the share of Evangelicals, with the effect being particularly pronounced

⁹As shown by Borusyak et al. (2022), in a shock-level SSIV setup, the F-statistic is calculated as the square of the t-statistic of the first-stage coefficient for the same sample used in the second-stage.

Table 2: **Migration and the Share of each Religious Group among Natives**

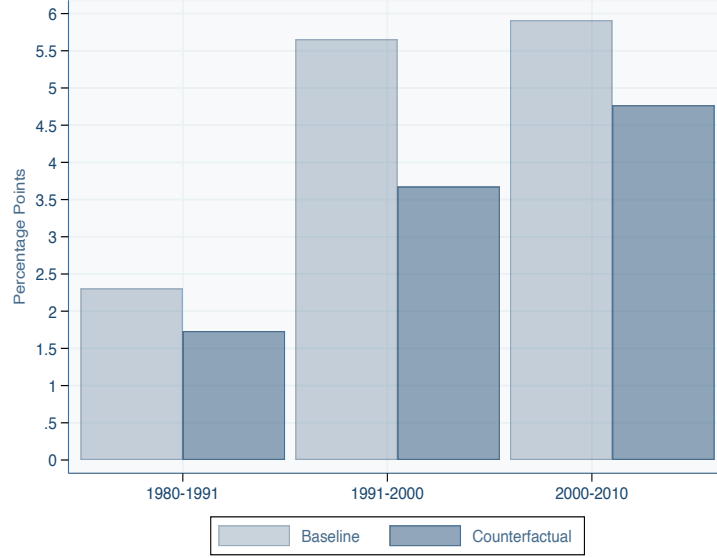
	Dep. Var.: Δ share of religious group			
	Catholics		Evangelicals	
	(1)	(2)	(3)	(4)
	OLS	SSIV	OLS	SSIV
<u>Panel A: Δ 1980-1991</u>				
In-migration rate	-0.023* (0.014)	-0.109* (0.063)	0.018** (0.007)	0.046* (0.025)
First Stage F Stat		186.4		181.6
Observations	3,919	3,989	3,919	3,989
<u>Panel B: Δ 1991-2000</u>				
In-migration rate	-0.050*** (0.011)	-0.167*** (0.020)	0.039*** (0.007)	0.148*** (0.030)
First Stage F Stat		199.1		186.6
Observations	3,919	3,989	3,919	3,989
<u>Panel C: Δ 2000-2010</u>				
In-migration rate	-0.032*** (0.005)	-0.103*** (0.014)	0.026*** (0.004)	0.073*** (0.010)
First Stage F Stat		69.0		68.4
Observations	3,919	3,989	3,919	3,989
Baseline covariates	✓	✓	✓	✓
Lagged religion share	✓	✓	✓	✓

This table shows municipality of destination-level OLS and municipality of origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level share of each religious group among native individuals. The SSIV is the outflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variation is the change in outcome between 1980 and 1991; in Panel B, between 1991 and 2000, while in Panel C, it is the change in the outcome between 2000 and 2010. All regressions include controls for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population) as well as the share of the respective religious group in the base year. Estimates are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

during the 1991–2000 period.

Finally, to provide a clearer perspective on the magnitude of the estimated effects, Figure 4 illustrates the variation in the share of Evangelicals across the three periods—1980–1991, 1991–2000, and 2000–2010—under both baseline and counterfactual scenarios in which migration rates are set to zero in all municipalities. The light-colored bars represent the average change in the Evangelical share across municipalities (i.e., the outcome variable in the regres-

Figure 4: **Average of Effects of Migration on the Share of Evangelical**



The light-colored bars represent the average change in Evangelical share across municipalities, weighted by each municipality's 1980 population, in 1980-1991, 1991-2000, and 2000-2010. The dark-colored bars depict the counterfactual scenario with zero migration rates. Formally, the counterfactual estimates are calculated as $\Delta s_d^{cf} = \Delta s_d - \hat{\beta} m_d$, where Δs_d denotes the change in the share of Evangelicals in municipality d during a given period, $\hat{\beta}$ is the (SSIV) point estimate from Table 2 for the corresponding period, and m_d represents the migration rate in municipality d . These values are then averaged over municipalities, weighted by their 1980 populations.

sions for Evangelicals), weighted by each municipality's 1980 population. The dark-colored bars show the same averages in a counterfactual scenario where migration rates are assumed to be zero. Formally, the counterfactual estimates are computed as $\Delta s_d^{cf} = \Delta s_d - \hat{\beta} m_d$, where Δs_d denotes the change in the share of Evangelicals in municipality d during a given period, $\hat{\beta}$ is the corresponding (SSIV) point estimate from Table 2, and m_d represents the migration rate in municipality d in that period. These values are then averaged across municipalities, weighted by their 1980 populations.

Back-of-the-envelope calculations based on Figure 4 indicate that migration has significantly influenced the growth of the Evangelical population in Brazil. For example, between 1991 and 2000, the average increase in the share of Evangelicals was approximately 5.5 percentage points in the baseline scenario, compared to 3.5 percentage points in a counterfactual scenario without migration. Cumulatively, over the three periods analyzed, the difference between these scenarios amounts to 3.7 percentage points, suggesting that the effects we are estimating are economically relevant.

3.5 Political Consequences

The institutional reallocation of religious market share documented above has profound implications for the distribution of political power. A distinct feature of the rise of Evangelicalism in Brazil—and globally—is its translation into political mobilization. Unlike the Catholic Church, which encompasses a diffuse and often ideologically heterogeneous population, Evangelical denominations have successfully organized into a cohesive voting bloc known as the Parliamentary Evangelical Front (*Frente Parlamentar Evangélica*, FPE).

Although migration may affect votes for Evangelical candidates through channels other than religious conversion, a vast literature has documented the close connection between the rise of Evangelicals and increasing electoral support for Evangelical politicians (Costa et al., 2019; Corbi and Sanches, 2024; Sola, 2024). In this light, it is natural to expect that the rise in Evangelical affiliation documented in Table 2 would be mirrored in higher vote shares for Evangelical candidates.

We test this hypothesis by estimating the effect of migration shocks on the electoral performance of Evangelical candidates in federal elections. We construct a dataset of vote shares for candidates formally affiliated with the FPE for the elections of 1998, 2002, 2006, and 2010.¹⁰ We pool data from these four electoral cycles and estimate our baseline SSIV specification, where the outcome variable is the municipality-level vote share for Evangelical candidates and the explanatory variable is the accumulated migration rate between elections, instrumented with the commodity-price shock.

Table 3 presents the results. Column 1 reports the specification without controls, while columns 2 through 4 progressively add municipality fixed effects, time fixed effects with baseline covariate interactions, and state-specific trends. Across all specifications, we find a positive, statistically significant, and economically large effect of migration inflows on the Evangelical vote share.

These results provide crucial validation for the depth of the religious transition. The fact that migration shocks reallocate not only self-reported religious affiliation (as seen in

¹⁰Data on Evangelical candidates was kindly provided by Fabio Lacerda and is described in Lacerda (2017). Members of the FPE for 1998-2010 are identified using FPE (2012); DIAP (2002); Dantas (2011); BRASIL (2003, 2018-2020). Electoral data are retrieved from TSE (2019).

Table 3: **Effects of Migration on the Vote Share of Evangelical Candidates**

	Dep. Var.: Evangelical vote share			
	(1)	(2)	(3)	(4)
In-migration rate	0.201*** (0.063)	0.253* (0.130)	0.232* (0.129)	0.148*** (0.053)
First Stage F Stat	420.8	224.1	210.4	474.3
Observations	15,956	15,956	15,956	15,956
Municipality FE		✓	✓	✓
Time FE		✓	✓	✓
Covariates			✓	✓
State trends				✓

This table shows origin-level SSIV coefficients on the vote share of Evangelical candidates. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. Columns 2-4 include municipality and election-year dummies, while Columns 3-4 also control for 1980 demographic covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population) interacted with the election-year dummies. Finally, column 4 also includes state-specific trends. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

the Census) but also voting behavior suggests that the expansion of Evangelical churches succeeds in deeply integrating new adherents into the denomination’s social and political network. The migration-induced market size shock, therefore, did not merely alter the religious landscape; it reshaped the political map of Brazil, fueling the rise of a conservative religious coalition consistent with the patterns of polarization observed in the recent political economy literature (Costa et al., 2019; Corbi and Sanches, 2024; Sola, 2024).

We interpret these electoral outcomes not as a separate mechanism to be modeled, but as a high-stakes validation of our main results—confirming that the migration-induced shift in religious affiliation represents a deep behavioral transformation with tangible power implications, rather than merely ‘cheap talk’ or superficial identity labeling.

3.6 Robustness Analysis

We assess the robustness of these results in several ways. First, concerns regarding the potential correlation between the instrument and unobserved factors affecting religious conversion appear to be limited in our context. Including controls for regional economic and demo-

graphic trends has minimal impact on the estimates, as shown in Tables B2 and B3—compare columns 1 (without controls) and 2 (with controls). In particular, the results remain stable when controlling for differential trends in key demographic characteristics, household income per capita, total population, and the lagged share of each religion.

Second, one might worry that destination food prices are correlated with commodity price shocks in the areas of origin. If poverty influences conversion, this correlation could threaten the validity of our identification. To address this concern, column 3 of both tables includes destination price shocks alongside the baseline. The estimates remain stable, with only minor changes observed for both Catholics and Evangelicals.

Third, we consider the possibility that price shocks in origin areas might affect destination economies through trade channels, as noted by Corbi et al. (2025), Imbert and Ulyssea (2022), and Kleemans and Magruder (2018). For example, destination municipalities may industrialize or strengthen commercial ties with origin regions. To account for this, we construct a measure of price shocks at the destination, weighted by the inverse distance to each origin. The rationale is that trade flows are generally stronger between municipalities that are closer to each other. Column 4 demonstrates that controlling for this measure does not alter our results.

Fourth, we investigate whether variations in religious composition across states affect our results. A comparison of columns 2 and 5, which includes state fixed effects, shows that our estimates are robust against this concern.

Fifth, we test the validity of the common pre-trend assumption. If municipalities exposed to migration shocks exhibit diverging pre-shock religious trends, our estimates might reflect those differences rather than a causal effect. Following Corbi et al. (2025) and Imbert and Ulyssea (2022), we incorporate the lagged dependent variable. Results in column 6 confirm that our main findings hold after accounting for these pre-existing trends.

Finally, we consider the persistence of migration. If migration flows are highly persistent, our instrument may reflect prior waves rather than current shocks (Jaeger et al., 2018). In column 7, we follow Imbert and Ulyssea (2022) and control for lagged migration. Our results remain stable across all specifications, and our baseline already accounts for population, which helps address long-term migration effects.

4 Two Channels of the Mechanism

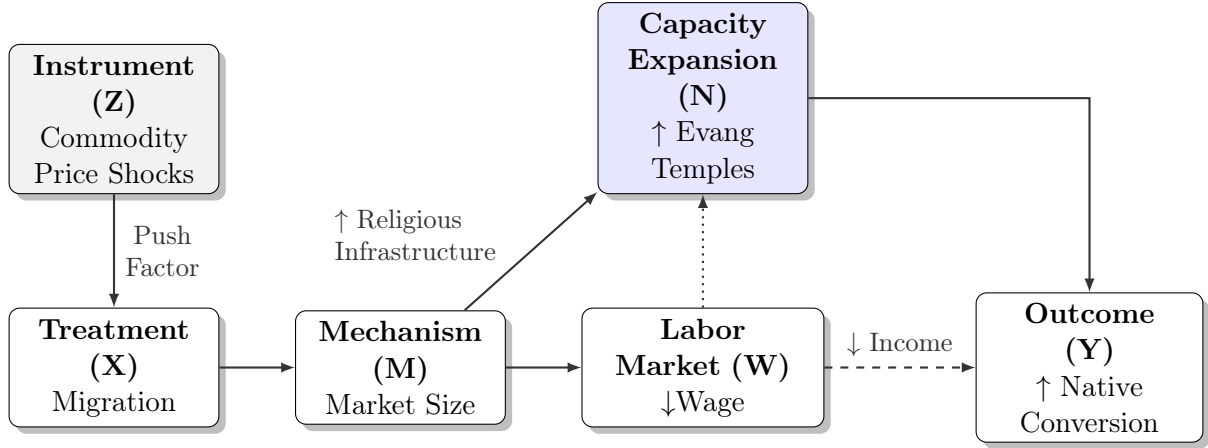
The reduced-form estimates in Section 3 show that migration-induced population shocks lead to a systematic decline in the Catholic share and a commensurate rise in the Evangelical share among native (non-migrant) residents. As we argued above, the observed religious change does not appear to stem from direct cultural transmission due to the *Composition Paradox*—migrants originated from predominantly Catholic regions, yet their arrival coincided with declining Catholic affiliation at destination. Rather, the transition seems to emerge from a more complex economic dynamic operating through changes in local market conditions.

In this section, we shed light on the economic mechanisms embedded in the reduced-form coefficient β of equation (1). We investigate two distinct channels through which migration affects religious affiliation, as illustrated by the schematic roadmap in Figure 5.

Income channel. As illustrated in the “Labor Market” node of Figure 5, the migration-induced expansion of market size (M) increases the effective supply of workers, putting downward pressure on wages. If religious demand exhibits differential income elasticities across denominations, these labor market shocks can induce predictable shifts in religious affiliation. A growing literature links economic insecurity to the rise of Evangelicalism in Brazil (Costa et al., 2019; Chesnut, 1997; Cavalcanti et al., 2022). Evangelical churches provide dense local networks, accessible pastoral attention, and immediate forms of material and emotional support—features that resemble the “club good” and social insurance mechanisms formalized in canonical models of group formation. In contrast, the Catholic Church operates with more hierarchical and centralized structures, offering less individualized support and slower adaptation to local shocks. If Evangelical churches provide stronger insurance against localized economic downturns, then negative income shocks should increase the relative demand for Evangelical affiliation.

Religious infrastructure channel. Migration also increases local market size. As shown by the solid path in Figure 5, the market-size shock (M) can trigger expansions of religious infrastructure (N). How strongly infrastructure responds to this shock depends, among other

Figure 5: **Causal Chain: From Commodity Shocks to Religious Change**



factors, on how affiliation responds to infrastructure expansion, a margin along which descriptive evidence suggests sharp differences between the Catholic Church and Evangelical denominations (see Section 2.2).

In what follows, we provide empirical evidence for both income and infrastructure channels, examining how migration affected local labor markets and how each denomination responded to the demographic shock through infrastructure expansion.

4.1 Income Channel

As illustrated in the ‘Labor Market’ node of Figure 5, the migration-induced expansion of market size (M) increases the effective supply of workers, putting downward pressure on wages.¹¹ If religious demand exhibits differential income elasticities across denominations, these labor market shocks can induce predictable shifts in religious affiliation. In this section, we analyze whether migration shocks induced labor market changes; the link between labor market and religious changes will be examined in the next section.

A growing literature links economic insecurity to the rise of Evangelicalism in Brazil (Costa et al., 2019). Evangelical churches provide dense local networks, accessible pastoral attention, and immediate forms of material and emotional support. These features resemble the ‘club good’ and social insurance mechanisms formalized in canonical models of group

¹¹Recent work on Brazilian internal migration documents sizable negative effects on wages and earnings (Imbert and Ulyssea, 2022; Corbi et al., 2025).

Table 4: **Effects of Migration on Earnings for Native Workers**

	(1)	(2)	(3)	(4)
	Overall	Formal	Informal	Family income
Panel A: OLS (Δ 1991-2000)				
In-migration rate	-0.019 (0.026)	0.009 (0.032)	-0.030 (0.031)	-0.018 (0.027)
Observations	3,919	3,919	3,919	3,919
Panel B: SSIV (Δ 1991-2000)				
In-migration rate	-0.401*** (0.092)	-0.217** (0.088)	-0.360*** (0.101)	-0.437*** (0.118)
First Stage F Stat	192.2	192.2	192.2	192.2
Observations	3,989	3,989	3,989	3,989
Panel C: OLS (Δ 2000-2010)				
In-migration rate	-0.012 (0.020)	-0.055** (0.026)	0.048** (0.020)	-0.004 (0.020)
Observations	3,919	3,919	3,919	3,919
Panel D: SSIV (Δ 2000-2010)				
In-migration rate	-0.540*** (0.199)	-0.435*** (0.149)	-0.618** (0.263)	-0.579*** (0.208)
First Stage F Stat	70.9	70.9	70.9	70.9
Observations	3,989	3,989	3,989	3,989
Baseline covariates	✓	✓	✓	✓

Each column reports destination-level OLS and origin-level SSIV estimates of changes in labor income or household income per capita. SSIV is constructed from predicted migration inflows. All regressions include lagged baseline covariates and are weighted by 1980 population. Standard errors clustered at the origin-municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

formation. In contrast, the Catholic Church operates with more hierarchical and centralized structures, offering less individualized support and slower adaptation to local shocks. If Evangelical churches provide stronger insurance against localized economic downturns, then negative income shocks should increase the relative demand for Evangelical affiliation.

We first examine whether migration shocks had consequences on labor income of native residents of Brazilian cities. We estimate variants of equation (1) where the dependent variable is the change in (log) earnings of native workers or per capita household income, instrumenting migration with the SSIV described in Section 3. Table 4 reports these results. Because labor market variables in the 1980 Census exhibit substantial non-response and inconsistencies, we focus on the 1991–2000 and 2000–2010 periods.

Across specifications, migration significantly depresses earnings of native workers, with the largest effects among informal workers. Back-of-the-envelope calculations indicate that, absent migration, average wages would have been 4.1% higher in 2000 and 6.3% higher in 2010. These findings are consistent with Corbi et al. (2025) and Imbert and Ulyssea (2022), who document similar magnitudes using different samples and variants of the SSIV approach.

Appendix Table B4 shows that the negative wage effects are robust to controlling for pre-trends, lagged migration, destination-level price shocks, trade-linkage weighted shocks, and state fixed effects. Only when controlling for lagged migration in 2000–2010 does significance fall, though point estimates remain economically meaningful.

Taken together, these results support the income mechanism: migration generates income shocks that increase the attractiveness of Evangelical churches relative to the Catholic Church, contributing to the native conversion patterns documented earlier.

4.2 Religious Infrastructure Channel

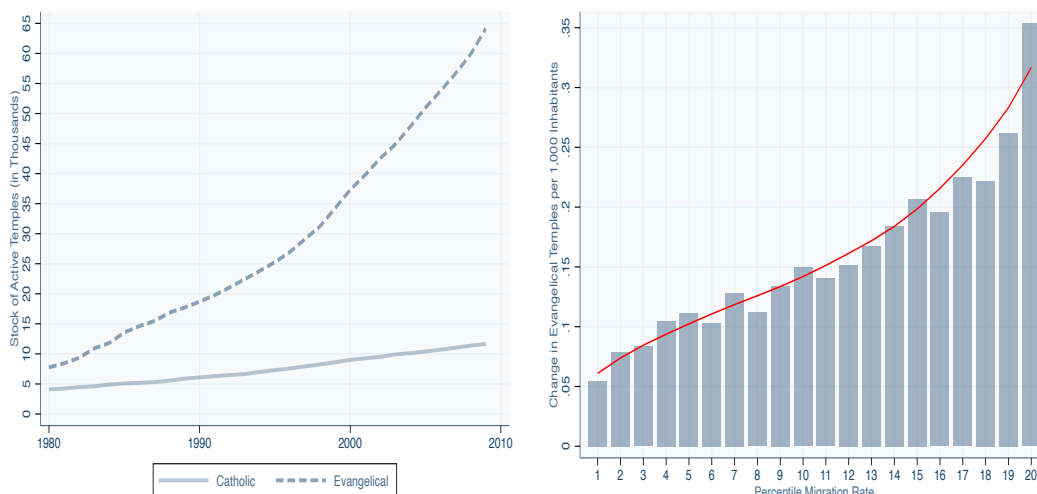
Migration also increases local market size. If demand for religious affiliation differs in its elasticity with respect to installed infrastructure, migration-induced market expansion can generate asymmetric infrastructure responses across denominations. As discussed in Section 2.2, descriptive evidence suggests that Catholic and Evangelical churches rely on markedly different conversion technologies: Evangelical churches appear substantially more effective at translating local presence into affiliation. This difference provides a natural explanation for why Evangelical infrastructure expanded so sharply—particularly in major migrant destination areas—while Catholic infrastructure remained comparatively stable.

Figure 6 provides descriptive evidence that corroborates these points. First, Panel (a) shows that while the number of Catholic temples doubled between 1980 and 2010, the number of active Evangelical temples increased almost sevenfold during the same period. In addition, Panel (b) reveals that this massive expansion of Evangelical temples was concentrated in cities with higher migration rates, perhaps suggesting that the Evangelical supply-side actively responded to migration shocks.

More formally, we analyze how Evangelical and Catholic churches respond to migration shocks by estimating model (1), using the change in the number of temples normalized by the

population in the baseline year in each municipality as the outcome variable. Put simply, we seek to determine whether Catholic and Evangelical churches expanded their religious infrastructure in response to migrant inflows. In our context, such a response would support the following mechanism: the arrival of migrant waves prompted churches to increase the supply of temples, which, in turn, may have also accommodated new native adherents.¹²

Figure 6: **Supply Side Behavior of Catholics and Evangelicals**



(a) Stock of temples (1980-2010)

(b) Migration and Pentecostal temples

Panel (a) shows the stock of active temples of the Catholic Church and Evangelical denominations in Brazil from 1980 until 2010. Data source: Internal Revenue Service. Panel (b) shows the relationship between (in)migration rates and the per capita growth of Evangelical temples across Brazilian municipalities between 1980-1991, 1991-2000, and 2000-2010. The x-axis represents percentiles of migration rates, ranging from the 1st (municipalities with the lowest migration rates) to the 20th (municipalities with the highest migration rates). The bars represent the average yearly variation (in units per 1000 inhabitants) of Pentecostal temples within each percentile, while the line represents the LOWESS smoothed series. Data source: Brazilian Census from 1980-2010 and Internal Revenue Service.

Table 5 provides evidence of the positive effect of migration on temple supply. Columns 1 and 2 present the OLS and SSIV estimates of migration on the number of Catholic temples, while columns 3 and 4 report the corresponding effects on Evangelical temples. Panels A, B, and C display these estimates for the 1980–1991, 1990–2000, and 2000–2010 periods, respectively. Coefficients and standard errors are estimated from the origin-level specification; standard errors are clustered at the origin municipality level. All regressions control for the same set of baseline covariates as those reported in Table 2 and are weighted by the total

¹²Migration may have affected temple supply through additional channels beyond direct responses to demographic growth. We discuss these alternative mechanisms later in this section and the next.

population in 1980.

From these results, we highlight the following key points. First, the effect of migration on the construction of Catholic temples is positive and statistically significant only in the first period. However, even in this case, the magnitude of the effect is substantially smaller than the corresponding estimate for Evangelical churches. Second, migration appears to have consistently contributed to the expansion of Evangelical temples across all periods. As in the previous section, SSIV estimates for Evangelical churches (Catholic Church) are generally larger (smaller) than the corresponding OLS estimates. This pattern may be driven by the tendency of migrants to settle in economically stronger regions, where the appeal of Evangelical churches may be relatively weaker and that of the Catholic Church is more pronounced.

The faster and more intense response of Evangelical churches to the market-size shock, compared to the Catholic Church, can reflect fundamental differences in their conversion technologies, specifically in how effective physical capital is at generating conversion. The results in Table 5 are consistent with these structural differences. We will explore this mechanism with more details in the next section.

We also compute back-of-the-envelope calculations to illustrate the impact of migration on the number of Evangelical temples per capita over the periods 1980–1991, 1991–2000, and 2000–2010. These calculations follow the same methodology used in Figure 4. The results suggest that, in the absence of migration, the number of temples per 1,000 inhabitants would be approximately 27% lower on average across years and municipalities—corresponding to declines of 38% in 1991, 25% in 2000, and 20% in 2010. These magnitudes highlight the relevance of this channel in accounting for the findings discussed in Section 3.

According to Figure 5, there are at least two mechanisms through which migration may have produced this substantial effect. First, by depressing local wages, migration may have shifted religious demand toward Evangelicalism; in equilibrium, churches would respond by increasing temple supply (the dotted arrow in Figure 5). Second, by increasing the local population, migration expanded the religious market, creating incentives to open more temples across the territory, serving both migrants and non-migrants alike (the arrow connecting “M” to “N” in Figure 5). The structural model developed in the next section allows us to

Table 5: **Effects of Migration on the Number of Temples of each Religious Group**

	Dep. Var.: Δ number of temples per capita			
	Catholics		Evangelicals	
	(1)	(2)	(3)	(4)
	OLS	SSIV	OLS	SSIV
<u>Panel A: Δ 1980-1991</u>				
In-migration rate	0.096*** (0.020)	0.079*** (0.016)	0.539*** (0.079)	0.883*** (0.151)
First Stage F Stat		185.9		185.9
Observations	3,919	3,989	3,919	3,989
<u>Panel B: Δ 1991-2000</u>				
In-migration rate	0.031*** (0.011)	-0.021 (0.015)	0.309*** (0.044)	0.685*** (0.128)
First Stage F Stat		192.2		192.2
Observations	3,919	3,989	3,919	3,989
<u>Panel C: Δ 2000-2010</u>				
In-migration rate	0.020*** (0.005)	-0.006 (0.014)	0.164*** (0.030)	0.651** (0.255)
First Stage F Stat		70.9		70.9
Observations	3,919	3,989	3,919	3,989
Baseline covariates	✓	✓	✓	✓

This table presents municipality of destination-level OLS and municipality of origin-level SSIV estimates of the effect of migration on the number of Catholic and Evangelical temples per inhabitant. Columns 1 and 2 show estimates for Catholic temples, while Columns 3 and 4 report estimates for Evangelical temples. The SSIV uses migration inflows predicted by price shocks at the origin. Panels A, B, and C display the 1980-1991, 1991-2000, and 2000-2010 results, respectively. All regressions include controls for lagged baseline covariates (share of population under 18 years old; share of non-white; share of female; share with high school education or higher; share of married individuals; household income per capita, and total population). Regressions are weighted by the total population in 1980. Standard errors are clustered at the origin municipality level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

assess the relative importance of these two explanations.

Tables B5 and B6 present robustness checks for the results reported in Table 5. These checks confirm that our main conclusions remain unchanged when we control for origin – destination correlations in price shocks (column 3), trade linkages (column 4), and a broad set of economic and demographic controls, including population composition, income levels, and state fixed effects (columns 1, 2, and 5). The main deviation arises when we include pre-trends (column 6) and lagged migration (column 7) as controls. In these cases, the

coefficient for Evangelicals (only) in the 2000–2010 period becomes less precise, although the point estimates remain positive and of meaningful magnitude.

5 Quantifying the Mechanisms

To shed light on the sources of the asymmetric infrastructure responses of Catholic and Evangelical churches to migration inflows, and to quantify the relative importance of the income and religious infrastructure channels in shaping Evangelical affiliation growth, this section develops and estimates a structural model of religious demand. We adopt a discrete choice framework in the spirit of Berry et al. (2004), in which individuals’ preferences over religious options depend on two key factors:¹³ their income—since the appeal of different religions may vary across income levels—and the local stock of temples for each religion, which captures the role of infrastructure availability in shaping religious choice.

We estimate the model and use it to conduct counterfactual simulations. While these exercises are not causal, they provide quantitative guidance on how much of the observed rise in Evangelical affiliation can plausibly be attributed to the income and infrastructure channels, and on why migration inflows generated asymmetric infrastructure responses across Catholic and Evangelical denominations. In particular, the model highlights differences in the elasticity of affiliation with respect to infrastructure as a key feature underlying this asymmetry.

5.1 Framework

The demand for religion focuses on how individuals substitute between the Catholic Church, Evangelical churches, and the outside option—i.e., no religion—depending on individuals’ and religions’ characteristics.¹⁴ To formalize this, we consider an individual i residing in municipality d in a given period.¹⁵ The individual’s objective is to choose a religion $j \in \chi =$

¹³See also Corbi et al. (2022) for an application of this approach to religious demand in the U.S.

¹⁴To keep the model estimation tractable, we exclude other religions. This group’s share is relatively small (on average, 1% of the population) and remains fairly stable over time. Moreover, the primary religious transition we analyze involves shifts from the Catholic Church to Evangelical denominations, which our model flexibly accommodates.

¹⁵For simplicity, we omit time notation from our models.

$\{0, C, E\}$ that maximizes her utility, where 0 represents the outside option, C the Catholic Church, and E an Evangelical denomination. The utility of individual i from choosing religion $j \neq 0$ in municipality d is given by:

$$u_{ijd} = \pi_j + \gamma_j \alpha_{id} + \delta_{jd} + \varepsilon_{ijd}, \quad (7)$$

where u_{ijd} is the total utility derived from religion j . The first term, π_j , represents the baseline preference for religion j that is common across individuals and municipalities. The second term, $\gamma_j \alpha_{id}$, introduces individual-level heterogeneity in preferences. Here, α_{id} captures how the taste for attributes of religion j , summarized by γ_j , varies with individual characteristics. The third term, δ_{jd} , captures the influence of religion-municipality specific factors, such as the availability of religious infrastructure in different markets (e.g., the number of temples), on the utility individuals derive from religion j . Finally, ε_{ijd} is an Extreme Value Type I error term, assumed to be iid across individuals, religious options, and municipalities. This assumption is standard in models of demand for differentiated products; see, for example, Berry and Haile (2021) for a complete overview of this literature.

The heterogeneous component of preferences, α_{id} , is modeled as:

$$\alpha_{id} = \alpha_0 + \alpha_1 w_{id}(m_d) + \sum_{l=1}^L \alpha_{2l} x_{lid} + \sigma \nu_{id}, \quad (8)$$

where α_0 denotes a baseline preference common to all individuals. The term $w_{id}(m_d)$ represents income, and α_1 captures how preferences for religion j vary with income. We write income as a function of migration rates to municipality d , m_d , to emphasize that it is affected by labor market fluctuations triggered by migration, as suggested by the estimates in Table 4. The summation $\sum_{l=1}^L \alpha_{2l} x_{lid}$ accounts for demographic characteristics x_{lid} —such as age, education, race, and gender—allowing for observed heterogeneity in preferences across individuals through the coefficients α_{2l} .¹⁶ Finally, ν_{id} captures unobserved individual hetero-

¹⁶We assume that education is not affected by migration. Education is costly and unlikely to respond to labor market shocks in the short run. As discussed below, our goal is to assess how migration influences the demand for religion through the two main channels identified in the previous section. The exercises remain informative even if other mechanisms also play a role.

geneity not explained by income or demographics. This specification thus allows preferences for religion to vary systematically with both observed and unobserved individual characteristics, and with income changes induced by migration-related labor market shocks.

We also note that the model does not explicitly account for inertia or for switching costs in religious choices; see Corbi et al. (2022). In our setting, this is unlikely to be a major concern, as most individuals were initially Catholic. Given this initial condition, religious heterogeneity was relatively low, and inertia in religious choices does not pose a significant threat to our analysis.

The component δ_{jd} captures how the presence of religious infrastructure in municipality d and other religion-municipality specific factors influence the utility of religion j . Specifically, we model δ_{jd} as:

$$\delta_{jd} = \delta_{0j} + \delta_{1j}N_{jd}(m_d) + \xi_{jd}, \quad (9)$$

where δ_{0j} represents a religion specific constant j , and $N_{jd}(m_d)$ denotes the number of temples of religion j in municipality d normalized by its population in 1980. The coefficient δ_{1j} captures the impact of religious infrastructure on utility. A greater availability of temples reduces the cost of accessing religious services and may enhance the visibility or perceived legitimacy of a religion within the community. The term ξ_{jd} accounts for unobserved, municipality-specific factors affecting the attractiveness of religion j .¹⁷ Churches may respond to demand shocks induced by migration waves by adjusting $N_{jd}(m_d)$ —as the results in Table 5 strongly suggest. We write it as a function of m_d to emphasize this channel. Through this process, we aim at inferring the sensitivity of the demand for religion to supply-side responses to migration shocks.

Substituting equations (8) and (9) into equation (7), we express the utility of individual

¹⁷Again, it is likely that this term also depends on migration. However, as we argued earlier, this does not affect the conclusions presented in this section.

i from choosing religion j in municipality d as:

$$u_{ijd} = \tilde{\delta}_{jd}(m_d) + \tilde{\gamma}_j w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{jl} x_{lid} + \tilde{\sigma}_j \nu_{id} + \varepsilon_{ijd}, \quad (10)$$

where $\tilde{\gamma}_j = \gamma_j \alpha_1$, $\tilde{\alpha}_{jl} = \gamma_j \alpha_{2l}$ and $\tilde{\sigma}_j = \gamma_j \sigma$ represent, respectively, the effect of income and demographic variables on the utility individuals derive from Church j . The term $\tilde{\delta}_{jd}(m_d)$ is defined as $\tilde{\delta}_{jd}(m_d) = \tilde{\delta}_{0j} + \delta_{1j} N_{jd}(m_d) + \xi_{jd}$ where, $\tilde{\delta}_{0j} = \pi_j + \gamma_j \alpha_0 + \delta_{j0}$. Therefore, $\tilde{\delta}_{jd}(m_d)$ absorbs Church and church-municipality effects, which include the number of temples and any unobserved municipality-level shock specific to religion j , on the utility for religion j .

The individual's choice set also includes an outside option, denoted by $j = 0$, which represents the non-religious, or secular, option. Following convention in the literature, we normalize the utility of the outside option as follows:

$$u_{i0d} = \varepsilon_{i0d}, \quad (11)$$

where ε_{i0d} is an Extreme Value Type I shock, assumed to be iid across individuals i and municipalities d . This assumption is innocuous in the sense that it only changes the interpretation of the estimates of the parameters in equation (10) from “levels” to “relative to the outside option” without influencing our counterfactuals.

Given these assumptions, if individuals choose the alternative $j \in \chi$ that maximizes their utility, then the demand for religion j of an individual residing in a municipality d can be characterized as:

$$S_{jd}(m_d, \mathbf{x}_{id}) = \begin{cases} \frac{\exp(\tilde{\delta}_{jd}(m_d) + \tilde{\gamma}_j w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{jl} x_{lid} + \tilde{\sigma}_j \nu_{id})}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} x_{lid} + \tilde{\sigma}_k \nu_{id})} & \text{if } j \in \chi \setminus 0, \\ \frac{1}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} x_{lid} + \tilde{\sigma}_k \nu_{id})} & \text{if } j = 0. \end{cases} \quad (12)$$

In equation (12) $\mathbf{x}_{id}$ represents the vector of individuals demographic characteristics. The aggregate demand for religion j in a municipality d is obtained by integrating equation (12) over the joint distribution of income and individual characteristics in each municipality:

$$\bar{S}_{jd}(m_d) = \begin{cases} \int \frac{\exp(\tilde{\delta}_{jd}(m_d) + \tilde{\gamma}_j w(m_d) + \sum_{l=1}^L \tilde{\alpha}_{jl} \mathbf{x}_l + \tilde{\sigma}_j \nu)}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} \mathbf{x}_l + \tilde{\sigma}_k \nu)} dF_d(w(m_d), \mathbf{x}, \nu) & \text{if } j \in \chi \setminus 0, \\ \int \frac{1}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} \mathbf{x}_l + \tilde{\sigma}_k \nu)} dF_d(w(m_d), \mathbf{x}, \nu) & \text{if } j = 0. \end{cases} \quad (13)$$

The joint cumulative distribution function $F_d(w(m_d), \mathbf{x}, \nu)$ captures the distribution of income $w(m_d)$ the demographic vector $\mathbf{x}$ and the unobserved individual characteristic within municipality d . By integrating over this distribution, the aggregate demand $\bar{S}_{jd}$ accounts for observed and unobserved heterogeneity in individual characteristics across municipalities.

5.2 Estimation and Results

We now outline the estimation of the structural model parameters, followed by a discussion of the baseline results.

Estimation. Estimation proceeds in two steps. In the first step, we estimate equation (12), that is, the parameters $\tilde{\delta}_{jd}$, $\tilde{\gamma}_j$, $\tilde{\alpha}_{jd}$, and $\tilde{\sigma}_j$ for all $j \neq 0$, l , and d . Following Berry et al. (2004), we assume that ν follows a standard normal distribution, independently and identically distributed across individuals and municipalities. Under this assumption, we estimate the first-stage parameters by Simulated Maximum Likelihood using individual-level data from the Brazilian Census.

Step 1 is computationally intensive due to the large number of observations and municipality-religion-specific effects, $\tilde{\delta}_{jd}$, that need to be estimated. To make the estimation feasible, we use a sample comprising 4% of the total microdata from the Brazilian Censuses of 1991, 2000, and 2010.¹⁸ Again, we do not use the 1980 sample due to the problems with the information on household income in that Census wave. Additionally, we excluded individuals younger than 16 years from the sample, as younger individuals are less likely to be actively making

¹⁸Using larger samples is infeasible because the computational time required to estimate the model grows exponentially with the number of observations. The selected sample is sufficiently large to provide robust estimates while keeping computation time manageable.

independent religious choices; instead, their religious affiliations may reflect those of their parents or caregivers. As before, the model is estimated separately for each year.

To estimate $\tilde{\delta}_{jd}$ within each municipality, a sufficiently large number of observations is necessary. Municipalities with fewer than 450 observations in the sample were excluded, as reducing this threshold often caused the models to fail to converge. After these adjustments, the sample for 1991 includes 188,978 observations in 181 municipalities; for 2000, it includes 257,335 observations in 249 municipalities; and for 2010, it includes 245,113 individuals in 305 municipalities.¹⁹ As a result, our sample focuses on the largest Brazilian municipalities, which is where the religious transition was concentrated.

The demand model estimation proceeds to Step 2, where we use the $\tilde{\delta}_{jd}$ to estimate $\tilde{\delta}_{jd}(m_d) = \tilde{\delta}_{0j} + \delta_{1j}N_{jd}(m_d) + \xi_{jd}$. The main challenge is the potential endogeneity of N_{jd} , which may be correlated with unobserved market-level factors, ξ_{jd} . Following Berry et al. (2004), separating demand estimation into two steps helps mitigate the impact of this issue on our results by ensuring that the estimates of $\tilde{\gamma}_j$ remain unaffected.

To address this concern, our baseline second-stage regression pools the three Census waves (1991, 2000, and 2010), interacts temple density with Census wave dummies, and includes both municipality–religion and year–religion fixed effects. Specifically, for each group j we estimate separately $\tilde{\delta}_{jdt} = \sum_t \delta_{1jt} N_{jdt} d_t + \psi_{jd} + \phi_{jt} + \eta_{jdt}$, where d_t s are dummies for the 1991, 2000 and 2010 Census waves, ψ_{jd} are municipality–religion fixed effects, ϕ_{jt} are year–religion fixed effects and η_{jdt} are transitory local demand shocks. Municipality fixed effects account for long-run differences in religious preferences driven by geography, history, and culture, while year fixed effects capture nationwide religious trends. Under this specification, identification of δ_{1j} relies on the assumption that church investments in infrastructure are uncorrelated with short-term fluctuations in unobserved church–municipality demand shifters around their long-term average, a premise we substantiate with additional evidence later in this section.²⁰

¹⁹The total number of observations in the 1991 sample is substantially smaller than in the other years. To account for these differences, all estimates will be weighted using individual (and municipality) Census weights specific to each year.

²⁰Although investments in temples (N_{jd}) may respond in part to short-term shifts in demand, they are largely shaped by persistent structural conditions, such as local historical, cultural, and economic factors. Building a temple requires planning, fundraising, land acquisition, and regulatory procedures, especially in the case of the Catholic Church, where infrastructure is more capital-intensive and hierarchical. Even

Next, we show our baseline estimates and then turn to counterfactual simulations that quantify the relative contributions of income and infrastructure channels to the migration-driven rise in Evangelical affiliation. In the concluding subsection, we undertake a comprehensive set of robustness exercises and bring additional evidence that lends further support to our identifying assumptions and to our main findings.

Baseline results. Tables C7, C8, and C9 in the Appendix report our key estimates. We estimate two versions of the model. The first includes demographic controls—education, age, and gender—as specified in the covariates x_{lid} in equation (8); the second omits these controls.

For conciseness, Table C7 reports only the key parameter $\tilde{\gamma}_j$ and the parameter $\tilde{\sigma}_j$ for Catholics and Evangelicals in each year. Standard errors, clustered at the municipality level, appear in parentheses. Panel A presents estimates from the model with observed demographic controls, while Panel B omits them. We first note that the term $\tilde{\sigma}_j$ is not statistically significant across all specifications, except for one case (Catholics in 1991 in the model with demographic controls). Even in that case, the coefficients attached to income are very similar to those obtained from a slightly modified version of the model estimated under the constraint $\tilde{\sigma}_j = 0$, as shown in Table C8. Given these patterns—and considering that estimating the model with unobserved individual heterogeneity, ν_{id} , is substantially more computationally demanding than the version without this component—we use the model in Table C8 as our baseline specification and conduct all subsequent exercises using this version.

It is nevertheless informative to note that across all specifications and years, in both models with (Table C7) and without (Table C8) the heterogeneity component $\tilde{\sigma}_j$, the estimates of $\tilde{\gamma}_j$ for Evangelicals are consistently negative and highly significant. This indicates that lower income is associated with a higher probability of choosing an Evangelical church. By contrast, estimates for the Catholic Church show no systematic pattern and are generally

among Evangelicals—who are organizationally more flexible—church entry requires coordination and capital, introducing some inertia. Crucially, these administrative and construction lags decouple the current stock of temples from transitory demand shocks. Because the realization of new infrastructure ($N_{jd,t}$) relies on decisions made at $t - k$, it is plausibly orthogonal to the contemporaneous error term $\eta_{jd,t}$ conditional on the fixed effects. Unlike prices in standard demand estimation models, which adjust quickly to transitory shocks, religious infrastructure tends to evolve more gradually, responding primarily to long-term shifts in market conditions.

not statistically significant, suggesting that income does not predict Catholic affiliation in a consistent manner.

Table C9 presents the second-stage estimates of δ_{1j} , using both Pooled OLS and Two-Way Fixed Effects (TWFE) specifications, for models with and without demographic controls in the first stage.²¹ Standard errors are clustered at the municipality level. In the Pooled OLS models, Evangelical temple supply is positively associated with demand in 1991 and 2000, though the estimates are imprecise and generally not statistically significant; the Catholic coefficients are larger and significant. In contrast, the TWFE models reveal a different pattern: the Catholic coefficients shrink and lose significance, while the Evangelical coefficients become larger and highly significant.

One notable feature of Table C9 is that the positive association between Evangelical temple density and the estimated demand shifter emerges clearly only after controlling for fixed effects. For the Catholic Church, the opposite occurs: a strong positive association in the OLS specification weakens substantially once fixed effects are introduced. This pattern is informative in evaluating potential endogeneity concerns. If unobserved demand shocks correlated with temple supply were driving the estimates, we would expect the coefficients to increase—rather than diminish—after controlling for fixed effects. Instead, we observe the reverse for Catholics and a pronounced strengthening for Evangelicals once long-run heterogeneity is absorbed, suggesting that omitted variable bias is unlikely to be the source of our main results. More broadly, this pattern mirrors the results in Tables 2 and 5, where SSIV estimates are consistently larger than OLS for Evangelicals, and the opposite holds for the Catholic Church. We return to this point later in the section and provide further evidence supporting the identifying assumptions behind our estimates.

Taken together, these findings indicate that the elasticity of Evangelical demand with respect to religious infrastructure is positive, whereas the corresponding elasticity for the Catholic Church is statistically indistinguishable from zero. They also show that religious demand responds through both income and infrastructure channels. The first result helps rationalize why Evangelical infrastructure responds so strongly to migration-induced market-

²¹Accordingly, Step 1 yields two sets of $\tilde{\delta}_{jd}$ estimates—one from the model with demographic controls and one without. In Step 2, we run separate regressions using each set of $\tilde{\delta}_{jd}$ values. As mentioned above, these estimates are based on the model imposing $\tilde{\sigma}_j = 0$; see Table C8.

size shocks, whereas Catholic infrastructure does not. Consistent with prior work, we find a negative income gradient among Evangelicals, with lower income associated with a higher likelihood of Evangelical affiliation (Costa et al., 2019; Chesnut, 1997). At the same time, Evangelical affiliation is strongly correlated with local temple availability. Combined with the evidence from the previous section—showing that migration led to income declines and a sharp expansion of Evangelical temples—these results support the view that migration altered the local environment in which individuals make religious choices through both income and infrastructure channels.

5.3 Counterfactuals

While the estimates in Tables C8 and C9 show that religious affiliation responds to both income and infrastructure, they do not directly quantify the relative importance of each channel. This section presents a set of counterfactual exercises designed to assess how much of the effect of migration on Evangelical growth—documented earlier in the paper—can be attributed to changes in income versus changes in religious infrastructure.

A central advantage of this structural approach is that it allows us to assess the relative magnitude of the income and religious infrastructure channels shaping religious demand. One might argue that because migration has a statistically modest effect on native wages in our reduced-form estimates, the prominence of the religious infrastructure channel would arise mechanically. This reasoning, however, conflates the size of the shock with the underlying preference parameters: even small wage changes could, in principle, generate large shifts in religious affiliation if the marginal utility of religious insurance were sufficiently high (Chen, 2010; Cavalcanti et al., 2022). By structurally recovering preference parameters, we evaluate this possibility directly.

Our approach is as follows. Using the demand model, we first simulate the effect of migration on the share of Evangelicals, under the assumption that migration influenced religious affiliation exclusively through changes in income and Evangelical temple supply.

Mathematically, this corresponds to computing the following difference:

$$\Delta \bar{S}_E = \frac{1}{\#(i, d, t)} \sum_{i, d, t} [S_{Edt}(w_{idt}, N_{Cdt}, N_{Edt}) - S_{Edt}(w_{idt}^o, N_{Cdt}^o, N_{Edt}^o)],$$

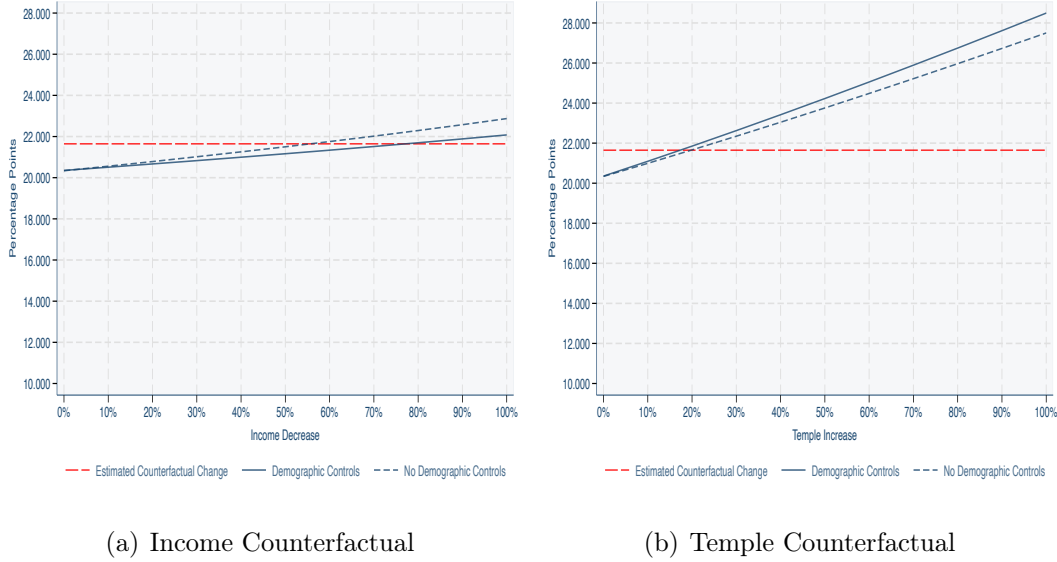
where $S_{Edt}(w_{idt}, N_{Cdt}, N_{Edt})$ denotes the share of Evangelicals in municipality d and year t predicted by the model using observed wages and temple stocks, and $S_{Edt}(w_{idt}^o, N_{Cdt}^o, N_{Edt}^o)$ is the predicted share under counterfactual (pre-migration) income, w_{idt}^o , and temple stocks, N_{Cdt}^o and N_{Edt}^o . The difference $\Delta \bar{S}_E$ represents the structural model analogue of the migration effect estimated in Section 3, under the assumption that migration affects religious affiliation only through the income and infrastructure channels.

We then scale pre-migration income, w_{idt}^o , downward uniformly, keeping temple stocks fixed at their pre-migration levels, until $\Delta \bar{S}_E = 0$. This yields the decline in income that would be required—on its own—to reproduce the effect of migration on the Evangelical share. We repeat the same procedure for pre-migration Evangelical temple supply, N_{Edt}^o , holding income fixed at pre-migration levels, to compute the change in infrastructure that would be necessary, by itself, to generate the same effect. Comparing these required changes with the actual migration-induced change obtained in the previous section allows us to assess which mechanism more plausibly accounts for the observed shift in affiliation. If reproducing the rise in Evangelical affiliation requires changes in income or infrastructure that far exceed the migration-induced changes estimated in the previous section, then those mechanisms are less likely to account for the observed effect. Conversely, if the required and estimated changes are of similar magnitude, the mechanism becomes a more plausible explanation.²²

The results of this exercise are presented in Figure 7. To construct each panel, we take the estimated demand model and evaluate it at pre-migration levels of income and temple supply for each individual in the sample. These pre-migration values— w_{idt}^o , N_{Cdt}^o , and N_{Edt}^o —are derived using the estimated effects of migration on income and infrastructure re-

²²As noted in the previous section, one limitation of this exercise is that it captures only the direct effect of income on religious demand, abstracting from potential indirect effects operating through temple supply. If lower income were to induce additional temple entry, and if that expansion in infrastructure increased religious affiliation, the role of income could be understated. This indirect mechanism corresponds to the dotted arrow in Figure 5. We return to this point in the next section and argue—based on the results presented here—that it is unlikely to overturn our conclusions.

Figure 7: **Changes in Income and Temple Supply Required to Rationalize the Observed Effects of Migration on the Share of Evangelicals**



Panel (a) shows the counterfactual Evangelical share under proportional income reductions, with $\varrho \in [0, 1]$, where $\varrho = 0$ denotes the baseline and $\varrho = 1$ eliminates income. Temple stock is fixed at its no-migration level. Panel (b) simulates proportional increases in Evangelical temples with income and Catholic temples held fixed at pre-migration levels. The dashed (solid) line refers to models estimated without (with) demographic controls. The horizontal line denotes the baseline Evangelical share S_E , and intersection points indicate the required change under each mechanism to rationalize the observed Evangelical share.

ported in Tables 4 and 5. Further details on their construction are provided in the Appendix. The upward-sloping curves in Figure 8(a) plot the average of $S_{Edt}((1 - \varrho_w)w_{idt}^o, N_{Cdt}^o, N_{Edt}^o)$ for different values of $\varrho_w \in [0, 1]$, while the curves in Figure 8(b) show the average of $S_{Edt}(w_{idt}^o, N_{Cdt}^o, (1 + \varrho_N)N_{Edt}^o)$ for different values of $\varrho_N \in [0, 1]$. Because income data are unavailable for 1980, and thus pre-migration income cannot be computed for 1991, all averages are calculated using only the 2000 and 2010 Census waves.

Each panel includes two curves—one based on the model estimated with demographic controls and one without—as well as a horizontal line indicating the average Evangelical share under observed conditions, i.e., the average of $S_{Edt}(w_{idt}, N_{Cdt}, N_{Edt})$.²³ The intersection between each curve and this horizontal line identifies the magnitude of change in income or temple supply required, on its own, to replicate the observed effect of migration on Evangelical affiliation.

Three patterns emerge. First, the difference between each curve and the horizontal line

²³For simplicity the horizontal line was computed using only predictions from the model estimated with demographic controls.

at $\varrho_w = 0$ and $\varrho_N = 0$ reflects the variation in the Evangelical share that would have occurred in the absence of migration as predicted by the demand model, assuming migration affected outcomes only through income and Evangelical temple supply. This gap is approximately 1.29 percentage points—close to the estimated impact of migration on the Evangelical share calculated from the estimates in Table 2 for 2000 and 2010, which is 1.53 percentage points; see also Figure 4.²⁴ This suggests that although migration may have influenced religious affiliation through additional channels, income and religious infrastructure were key mechanisms.

Second, the Evangelical share is substantially more responsive to temple supply than to income: the curves in the right panel are clearly steeper, indicating stronger sensitivity to changes in infrastructure.²⁵

Third, the size of the income drop or the temple expansion needed to reproduce the observed Evangelical share varies substantially between the two channels. Matching the observed Evangelical share through income alone would require a drop of 60–80%—an implausibly large decline compared to the 5% impact estimated in Table 4. By contrast, a 20% increase in Evangelical temples is sufficient to replicate the shift, closely matching the estimated effect of migration on infrastructure we estimated in the previous section.

Taken together, these exercises suggest that religious demand is substantially more responsive to infrastructure than to income. While Evangelical churches do appear to attract poorer individuals—consistent with prior literature (Chesnut, 1997; Mariano, 2011; Costa et al., 2019)—the magnitude of that association is modest. Our results suggest that in the context of rapid demographic growth, physical access (supply) acts as a binding constraint that overrides pure income preferences.

5.4 Robustness and Supporting Evidence

The results above rely on identifying assumptions that merit further scrutiny. The objective of this section is to assess the credibility of these assumptions and provide additional support

²⁴Recall that the counterfactual simulations are based only on the 2000 and 2010 Census waves, so we compare them to the migration effects for those years reported in Section 3.

²⁵Figure F3 in the Appendix provides additional sensitivity analysis supporting this conclusion.

for our main findings.

A first-order concern is that the stronger sensitivity of demand to changes in infrastructure may be driven by the endogeneity of infrastructure in the second step of our estimation procedure. First, as previously noted, the comparison between OLS and TWFE estimates in Table C9 suggests that endogeneity is unlikely to be driving the positive association between Evangelical temple density and demand. Including fixed effects—a standard approach to address unobserved heterogeneity—makes the Evangelical coefficient larger and more significant. If our estimates were upward-biased due to omitted variables correlated with both temple supply and religious demand, we would expect the opposite pattern. We interpret this as strong evidence that the estimated sensitivity of demand to infrastructure is not primarily driven by concerns of this nature.

A complementary piece of evidence on the association between Evangelical infrastructure and affiliation comes from a staggered difference-in-differences design that exploits the timing of temple entry across municipalities to further assess the validity of the estimates in Table C9. This exercise estimates the effect of temple entry directly on religious shares, rather than on the estimated demand shifters.²⁶ We find that the entry of Evangelical temples generates a positive and increasing effect on Evangelical affiliation, with no evidence of pre-trends (see Figure D2 in the Appendix). In contrast, the estimated effects of Catholic temple entry are positive but very imprecise and statistically insignificant. Using a different sample and identification strategy, these results corroborate our baseline estimates: Evangelical infrastructure expansion is systematically associated with rising Evangelical affiliation, while Catholic temple entry appears unrelated to changes in Catholic shares. This reinforces the interpretation that the relationship documented in Table C9 is not driven by unobserved confounders.

Additionally, we examine the dynamic response of temple supply to migration. Figure D1 in the Appendix reports SSIV estimates of the effect of migration shocks on temple openings over horizons $k = 0, \dots, 9$. For each k , we regress $N_{jd,t+k} - N_{jd,t+k-1}$ on m_{dt} , pooling across

²⁶Due to the one-to-one mapping between demand shifters and market shares, the results are qualitatively comparable to those in Table C9. The sample includes only municipalities with no temples of a given denomination in 1980. For estimation, we follow Callaway and Sant’Anna (2021). See Equation (D1) in the Appendix.

Census waves, and including the same controls as in Table 5 plus year fixed effects. The coefficients trace the impact of a migration shock at t on temple growth k periods ahead. It shows that migration increases Evangelical temple density over time, but with a lag—effects only become substantial after five years. In contrast, the impact on Catholic temples is negligible at all horizons. This pattern indicates that infrastructure responds relatively slowly to demographic shocks, and that any contemporaneous correlation between temple counts and short-run demand shocks is likely to be second-order.²⁷

A separate concern is that the weaker relationship between income and religious demand may also be driven by income endogeneity in the first step. If income is correlated with unobserved individual characteristics—such as family background or cultural traits—that also influence religious affiliation, the estimated coefficients $\tilde{\gamma}_j$ may be biased. While we cannot fully rule out this possibility, we assess its plausibility by examining how sensitive the estimated income gradient is to the inclusion of demographic controls. Figure 8(a) shows that controlling for observables—such as education, age, and gender—reduces the estimated sensitivity of demand to income: the dashed curve (estimated without controls) lies consistently above the solid curve (with controls), indicating a weaker association once basic demographics are taken into account. If income endogeneity were the main driver of the observed relationship, we would expect the inclusion of controls to strengthen—not weaken—the negative gradient. This pattern suggests that endogeneity is unlikely to explain the limited role of income in accounting for the rise in Evangelical affiliation.

Finally, one may also argue that the supply of Evangelical temples is itself a function of income: changes in income may shift religious demand, which in turn could trigger a supply response. In this view, a decline in w would increase the attractiveness of Evangelical churches, leading to an expansion in N_E . As a result, the total effect of lower income on Evangelical affiliation could be stronger than what is shown in Figure 8(a), which isolates only the direct effect of w . While we admit that this concern is valid, our results strongly suggest that the demand for Evangelical churches is not very sensitive to income. This limited responsiveness implies that any indirect effect of w on N_E —and thus on religious

²⁷A potential problem here is that delayed temple expansion may simply reflect persistence in migration flows. However, we show that in general our results remain stable when controlling for lagged migration, which tends to attenuate this concern.

affiliation—is likely to be small.²⁸

6 Conclusion

This paper shows that migration-induced expansions in market size can generate large reallocations of market shares when organizations differ in how effectively expanded capacity translates into participation. In the Brazilian religious market, internal migration did not reshape affiliation through the importation of religious preferences. Instead, we document a *Composition Paradox*: although migrants originated disproportionately from Catholic strongholds, their arrival reduced Catholic affiliation and increased Evangelical affiliation among natives, ruling out cultural transmission as the primary driver of religious change.

We show that this transformation operated through a market-size mechanism. Population inflows expanded local religious markets and triggered asymmetric expansion of religious capacity. Because affiliation responds much more strongly to Evangelical infrastructure than to Catholic infrastructure, this expansion translated into large gains in Evangelical market share. A structural model confirms that differences in the elasticity of affiliation with respect to capacity explain both the uneven infrastructure response and the resulting reallocation of affiliation.

More broadly, the findings highlight a general mechanism through which market-size shocks can reshape market structure even when preferences remain stable: demographic growth favors organizations whose technologies are better suited to converting expanded capacity into participation.

²⁸This concern stems from the fact that we do not explicitly model the supply side of religion. Doing so would require strong assumptions and pose significant challenges. First, defining geographic markets is particularly difficult in large metropolitan areas like São Paulo and Rio de Janeiro. Second, the number of religious organizations is vast and highly heterogeneous, complicating the structure and solution of any equilibrium model. Third, the objective function of churches may not be well-defined or easily modeled. These limitations prevent a precise understanding of the origins of the observed supply-side response, which we interpret—given the limited role of income in shaping demand—as a consequence of population growth induced by migration. While this interpretation is consistent with the reduced-form evidence, it is not derived directly from the model.

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Appendix

This appendix provides supporting material and robustness checks for the main findings in the paper. Section A documents the institutional background and the classification of religious denominations in Brazil used throughout the analysis. Section B presents robustness checks for the reduced-form SSIV results. Section C reports the structural estimates for the determinants of religious demand. Section D includes additional figures and tables illustrating dynamic and difference-in-differences supply-side effects. Section E provides technical details on how we construct the counterfactual simulations reported in the main text. Section F provides additional sensitivity analysis of religious demand with respect to changes in income and temple supply.

A Institutional Background Details

In this appendix, we provide a detailed explanation of the classification of religious denominations in Brazil, particularly the composition of the Evangelical group as used in this study. Table A1 presents the main religious traditions and the denominations within each category. This classification builds upon prior work, notably Mariano (2014), and offers a structured taxonomy for analyzing religious affiliation in Brazil.

The largest religious tradition in Brazil is the Catholic Church (Group I), which primarily consists of the Roman Catholic Church. Although the Orthodox Church is present in Brazil, its adherents are relatively few and do not significantly impact the broader religious landscape.

The Evangelical denominations are categorized into three distinct groups: Mainline Protestant (II), Non-Pentecostal Evangelical (III), and Pentecostal Evangelical (IV), with a separate category for Non-denominational Evangelicals (V). The classification of Evangelical denominations is not always clear-cut, as theological and institutional differences can blur traditional boundaries. Nonetheless, this taxonomy provides a useful framework for distinguishing major branches.

Mainline Protestant churches (II) include Lutheran, Anglican, Calvinist, and Anabaptist traditions. These groups are historically connected to the Protestant Reformation and were introduced to Brazil primarily through European immigration. These churches tend to have more hierarchical structures and formalized doctrines compared to other Evangelical groups.

Non-Pentecostal Evangelicals (III) include denominations such as Presbyterians, Congregationalists, Baptists, Methodists, and Adventists. Many of these groups were influenced by the religious revivals of the 18th and 19th centuries, particularly the Second Great Awakening in the United States. They emphasize personal conversion experiences, scriptural authority, and active missionary work.

Pentecostal Evangelicals (IV) encompass a broad array of denominations, many of which emerged in Brazil during the 20th century. These churches emphasize direct experiences with the Holy Spirit, such as speaking in tongues, divine healing, and prophecy. They are often further divided into Classical Pentecostals, Neo-Pentecostals, and newer Charismatic movements. Notable denominations in this category include the Assemblies of God, Universal Church of the Kingdom of God, and the Foursquare Church. Neo-Pentecostalism,

in particular, is characterized by its embrace of the prosperity gospel and highly dynamic worship services (Zilla, 2018).

Table A1: **Religious Denominations in Brazil**

	Group	Denominations
I	Catholic	Roman Catholic
II	Mainline Protestant	Lutheran, Anglican, Calvinist, Anabaptist
III	Non-Pentecostal Evangelical	Presbyterian, Congregationalist, Baptist, Methodist, Adventist
IV	Pentecostal Evangelical	Christian Congregation, Assembly of God, Foursquare, Universal Church of the Kingdom of God, Brasil para Cristo, Deus é Amor, Renascer, Mundial, Nazareno, Casa da Benção, Casa da Oração, Maranata, Igreja da Graça
V	Non-denominational Evangelicals	Various minor Evangelical denominations following hybrid (Pentecostal and Non-Pentecostal) lines
VI	Other	Afro-Brazilian, Spiritism, Eastern Religions, Judaism, Islam

Non-denominational Evangelicals (V) include various independent churches that do not align strictly with Pentecostal or traditional Protestant categories. These groups often blend elements from different theological traditions and exhibit a high degree of organizational flexibility. Their growth has been fueled by urbanization and the increasing individualization of religious practice.

Lastly, the category “Other” includes religious traditions outside of Christianity, such as Afro-Brazilian religions (e.g., Candomblé and Umbanda), Spiritism, Eastern religions, Judaism, and Islam. These groups, while historically significant, represent a relatively small fraction of the Brazilian population.

B Robustness Checks

This Appendix presents a comprehensive set of robustness checks for our reduced-form SSIV estimates. We examine the effects of migration on Catholic and Evangelical shares (Tables B2 and B3), on labor market outcomes (Table B4), and on the number of temples for each religious group (Tables B5 and B6). These results verify that our findings are robust to a range of controls and specifications.

Table B2: **Effects of Migration on the Share of Catholics among Native Populations**

	Dep. Var.: Δ share of catholics						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	-0.118* (0.069)	-0.109* (0.063)	-0.108* (0.063)	-0.109* (0.063)	-0.121*** (0.030)		
First Stage F Stat	185.9	186.4	176.6	186.4	343.9		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	-0.181*** (0.026)	-0.167*** (0.020)	-0.166*** (0.019)	-0.167*** (0.020)	-0.122*** (0.018)	-0.166*** (0.020)	-0.081*** (0.022)
First Stage F Stat	192.2	199.1	189.7	210.5	267.6	183.0	103.6
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	-0.120*** (0.018)	-0.103*** (0.014)	-0.103*** (0.014)	-0.103*** (0.014)	-0.078*** (0.015)	-0.093*** (0.014)	-0.042*** (0.010)
First Stage F Stat	70.9	69.0	68.6	69.5	160.4	64.6	58.7
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	✓
Lagged share of catholics		✓	✓	✓	✓	✓	✓
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level share of catholics. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C is the change between 1991-2000 and 2000-2010, respectively. Columns 2-7 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population) as well as the lagged share of Catholics. Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B3: **Effects of Migration on the Share of Evangelicals among Native Populations**

	Dep. Var.: Δ share of evangelicals						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	0.041* (0.023)	0.046* (0.025)	0.047* (0.026)	0.047* (0.025)	0.043*** (0.011)		
First Stage F Stat	185.9	181.6	171.9	181.2	347.5		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	0.152*** (0.031)	0.148*** (0.030)	0.146*** (0.030)	0.144*** (0.030)	0.098*** (0.015)	0.143*** (0.029)	0.060* (0.034)
First Stage F Stat	192.2	186.6	176.2	193.6	280.7	182.5	106.1
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	0.083*** (0.012)	0.073*** (0.010)	0.072*** (0.009)	0.073*** (0.009)	0.052*** (0.011)	0.041*** (0.008)	0.025*** (0.007)
First Stage F Stat	70.9	68.4	68.1	68.7	173.6	54.3	61.2
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	✓
Lagged share of evangelicals		✓	✓	✓	✓	✓	✓
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level share of evangelicals. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C is the change between 1991-2000 and 2000-2010, respectively. Columns 2-7 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population) as well as the lagged share of Evangelicals. Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B4: **Effects of Migration on Earnings of Native Workers**

	Dep. Var.: Δ log-earnings						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1991-2000</u>							
In-migration rate	-0.344*** (0.095)	-0.401*** (0.092)	-0.404*** (0.093)	-0.396*** (0.092)	-0.303*** (0.051)	-0.300*** (0.070)	-0.296** (0.135)
First Stage F Stat	406.5	192.2	184.0	198.7	297.1	173.1	107.5
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel B: Δ 2000-2010</u>							
In-migration rate	-0.402*** (0.126)	-0.540*** (0.199)	-0.540*** (0.197)	-0.540*** (0.199)	-0.256*** (0.092)	-0.628*** (0.237)	-0.270 (0.197)
First Stage F Stat	227.7	70.9	70.9	71.1	191.9	73.3	63.0
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	
Lagged share of evangelicals		✓	✓	✓	✓	✓	
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level log-earnings for native workers. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1991-2000, while in Panel B is the change between 2000-2010. Columns 2-6 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population). Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B5: **Effects of Migration on the Number of Catholic Temples**

	Dep. Var.: Δ number of catholic temples						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	0.068*** (0.013)	0.079*** (0.016)	0.076*** (0.016)	0.079*** (0.016)	0.070*** (0.013)		
First Stage F Stat	326.8	185.9	177.7	185.2	366.9		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	-0.002 (0.010)	-0.021 (0.015)	-0.023 (0.016)	-0.021 (0.015)	-0.057*** (0.016)	-0.018 (0.016)	0.012 (0.020)
First Stage F Stat	406.5	192.2	184.0	198.7	297.1	194.5	107.5
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	-0.003 (0.009)	-0.006 (0.014)	-0.007 (0.014)	-0.006 (0.014)	0.010 (0.010)	-0.006 (0.014)	0.033** (0.013)
First Stage F Stat	227.7	70.9	70.9	71.1	191.9	76.6	63.0
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level number of catholic temples. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C is the change between 1991-2000 and 2000-2010, respectively. Columns 2-6 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population). Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses.

*** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B6: **Effects of Migration on the Number of Evangelical Temples**

	Dep. Var.: Δ number of evangelical temples						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	0.863*** (0.117)	0.883*** (0.151)	0.882*** (0.151)	0.883*** (0.151)	0.858*** (0.091)		
First Stage F Stat	326.8	185.9	177.7	185.2	366.9		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	0.748*** (0.115)	0.685*** (0.128)	0.686*** (0.129)	0.688*** (0.128)	0.565*** (0.077)	0.538*** (0.156)	0.419*** (0.160)
First Stage F Stat	406.5	192.2	184.0	198.7	297.1	88.2	107.5
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	0.737*** (0.178)	0.651** (0.255)	0.654*** (0.247)	0.651** (0.255)	0.516*** (0.101)	0.360 (0.230)	0.284 (0.227)
First Stage F Stat	227.7	70.9	70.9	71.1	191.9	34.9	63.0
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level number of evangelical temples. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C, it is the change between 1991-2000 and 2000-2010, respectively. Columns 2-6 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population). Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

C Structural Model Estimates

This Appendix reports the structural estimates of our demand model. Tables C7 and C8 present the estimated coefficients on income in the demand for Catholic and Evangelical affiliation for models with and without unobserved individual heterogeneity (term ν_{id} in the demand model), respectively. Table C9 reports the relationship between the number of temples and the estimated demand shifters $\tilde{\delta}_{jd}$ across years and specifications, using both OLS and TWFE estimators.

Table C7: **First-Stage Estimates of Preference Heterogeneity by Religion and Year (Mixed Logit)**

	(1)	(2)	(3)	(4)	(5)	(6)
	1991		2000		2010	
	Catholic	Evangelical	Catholic	Evangelical	Catholic	Evangelical
Panel A: With Demographic Controls						
Income	-0.0214*** (0.0007)	-0.1616*** (0.0010)	0.0099*** (0.0004)	-0.1204*** (0.0006)	-0.0024*** (0.0001)	-0.1128*** (0.0002)
ν	0.1928** (0.0673)	-0.0375 (0.0688)	0.0198 (0.0674)	-0.0250 (0.0490)	0.0108 (0.0447)	-0.0215 (0.0307)
Observations	1,735,871	1,735,871	2,481,561	2,481,561	3,123,037	3,123,037
Panel B: Without Demographic Controls						
Income	-0.0048*** (0.0006)	-0.1664*** (0.0009)	0.0708*** (0.0004)	-0.1094*** (0.0005)	0.0110*** (0.0001)	-0.1436*** (0.0002)
ν	0.0277 (0.0894)	-0.0988 (0.0678)	0.0079 (0.0674)	-0.0238 (0.0488)	0.0020 (0.0445)	-0.0168 (0.0306)
Observations	1,735,871	1,735,871	2,301,341	2,301,341	2,956,642	2,956,642
Municipality Dummies	✓	✓	✓	✓	✓	✓

This table presents estimates of the effect of income and of unobserved individual heterogeneity (ν) on religious demand, following equation (12). The model is estimated using Simulated Maximum Likelihood for samples of individuals from the Brazilian Census for the corresponding years. Individuals' choice sets include three options: Catholic, Evangelical, and no religion. Panel A includes demographic controls (education, age profile, and gender dummies), while Panel B does not. All regressions include municipality fixed effects. Standard errors are clustered at the municipality level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table C8: **First-Stage Estimates of Preference Heterogeneity by Religion and Year (Conditional Logit)**

	(1)	(2)	(3)	(4)	(5)	(6)
	1991		2000		2010	
	Catholic	Evangelical	Catholic	Evangelical	Catholic	Evangelical
<u>Panel A: With Demographic Controls</u>						
Income	-0.00485 (0.00845)	-0.166*** (0.0229)	0.0708*** (0.0158)	-0.109*** (0.0215)	0.0110 (0.0103)	-0.144*** (0.0134)
Observations	1,735,871	1,735,871	2,481,561	2,481,561	3,123,037	3,123,037
<u>Panel B: Without Demographic Controls</u>						
Income	-0.0214*** (0.00649)	-0.161*** (0.0249)	0.00696 (0.00559)	-0.120*** (0.0163)	-0.00142 (0.00201)	-0.106*** (0.0109)
Observations	1,735,871	1,735,871	2,301,341	2,301,341	2,956,642	2,956,642
Municipality Dummies	✓	✓	✓	✓	✓	✓

This table presents estimates of the effect of income on religious demand, following equation (12). The model is estimated using Maximum Likelihood for samples of individuals from the Brazilian Census for the corresponding years. Individuals' choice sets include three options: Catholic, Evangelical, and no religion. Panel A includes demographic controls (education, age profile, and gender dummies), while Panel B does not. All regressions include municipality fixed effects. Standard errors are clustered at the municipality level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table C9: Effects of Number of Temples on the Demand Shifters $\tilde{\delta}_{jd}$

	(1)	(2)	(3)	(4)
	No Demographic Controls		Demographic Controls	
	Evangelical	Catholic	Evangelical	Catholic
<u>Panel A: Pooled OLS</u>				
Number of Temples $\times$ Year = 1991	0.264 (0.419)	7.097** (2.828)	0.170 (0.405)	6.783** (2.752)
Number of Temples $\times$ Year = 2000	0.000749 (0.234)	6.114*** (1.083)	0.0855 (0.224)	5.720*** (0.976)
Number of Temples $\times$ Year = 2010	-0.208 (0.197)	8.294*** (0.840)	-0.150 (0.186)	8.059*** (0.811)
<u>Panel B: TWFE</u>				
Number of Temples $\times$ Year = 1991	0.916*** (0.322)	1.926 (1.957)	0.792** (0.316)	1.858 (1.957)
Number of Temples $\times$ Year = 2000	0.926*** (0.310)	-1.368* (0.786)	0.916*** (0.306)	-1.510* (0.820)
Number of Temples $\times$ Year = 2010	0.726*** (0.232)	-0.754 (0.781)	0.688*** (0.229)	-0.717 (0.777)
Observations	735	735	671	671

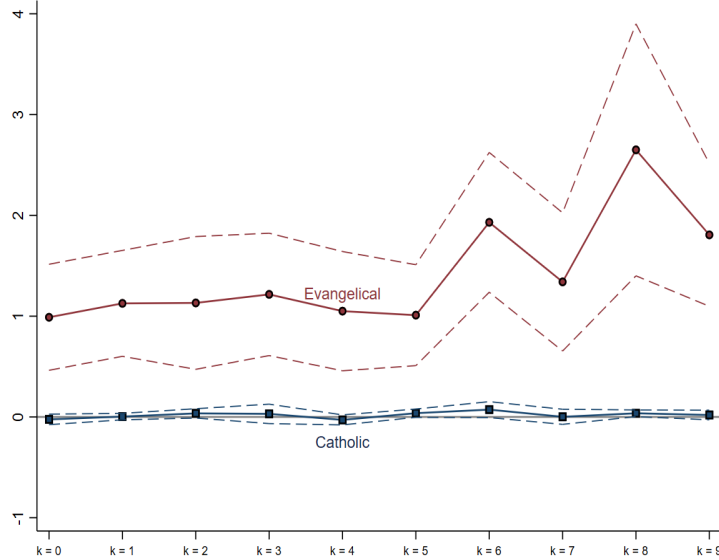
This table reports pooled OLS and two-way fixed effects (TWFE) estimates of the effect of the number of temples on the demand shifters, $\tilde{\delta}_{jd}$, for each religious group. All regressions include year fixed effects, and TWFE specifications additionally include municipality fixed effects. Regressions are estimated separately for each group. The coefficients on the number of temples are interacted with year dummies. In models with demographic controls, the dependent variable $\tilde{\delta}_{jd}$ is obtained from a first-stage regression that includes income, education, age profile, and gender dummies. Models without demographic controls exclude education, age profile, and gender dummies. Standard errors, clustered at the municipality level, are reported in parentheses.

D Additional Tables and Figures

This Appendix presents additional graphical evidence on the dynamic effects of migration and temple entry. Figure D1 illustrates how migration shocks affect the entry of temples over time, while Figure D2 reports staggered difference-in-differences estimates of the effect of temple entry on religious affiliation.

Dynamic effects of migration. The exercise in Figure D1 is based on SSIV regressions of $N_{jd,t+k} - N_{jd,t+k-1}$ on m_{dt} , with $k \in 0, 1, 2, \dots, 9$. For each value of k , a separate regression is estimated for each religious group, pooling data across all Census years. Intuitively, the coefficient on m_{dt} for a given k captures the average effect—across municipalities and Census waves—of a migration shock at time t on the change in the stock of temples of each religious group k periods ahead. All regressions include the same baseline controls used in Table 5, along with year fixed effects. The coefficients plotted in Figure D1 represent the estimated impact of m_{dt} on $N_{jd,t+k} - N_{jd,t+k-1}$ for each horizon k . The results suggest that migration has a stronger effect on investments in Evangelical temples in the longer run. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Figure D1: **Effects of migration on the dynamic entry of temples**



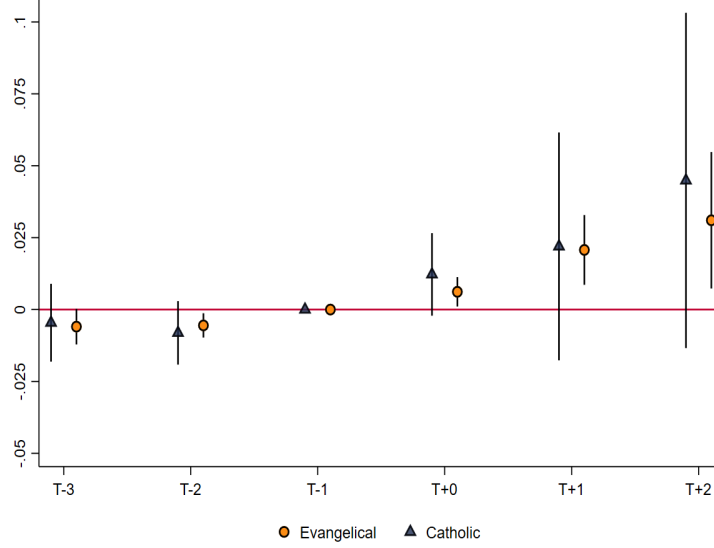
This figure shows the dynamic effects of in-migration on the entry of Catholic and Evangelical temples. Each point is an SSIV coefficient for a regression of a contemporaneous shock of incoming migration on changes in the number of temples per capita k years ahead. Dashed lines represent the 95% confidence interval.

Differences-in-Differences for temple entry. The results in Figure D2 are based on the estimation of the following equation:

$$Y_{it} = \alpha_i + \beta_t + \sum_{\substack{h=-a \\ h \neq -1}} \tau_h \mathbb{1}[K_{it} = h] + \tau_b \mathbb{1}[K_{it} \geq b] + \varepsilon_{it}, \quad (\text{D1})$$

where, α_i and β_t are municipality and time fixed effects; $a \geq 0$ and $b \geq 0$ are the included leads and lags of the treatment, and ε_{it} is the error term. We define a municipality as treated if, between Census years t and $t + 1$, at least one temple began operating.

Figure D2: **Effects of the Dynamic Entry of Temples on Religious Affiliation**



This figure shows the dynamic effects of temple entry on the proportion of the population affiliated with each religious group, using the estimator developed in Callaway and Sant’Anna (2021). For each religious group, the sample includes only municipalities with no temples of that denomination operating in 1980. There are 2,445 municipalities without Evangelical temples in 1980 and 2,100 without Catholic temples, corresponding to 61.7% and 53.0% of the original sample, respectively. Vertical lines on each coefficient represent 95% confidence intervals.

E Counterfactual Details

This Appendix provides full details on the construction of counterfactual variables used in our simulations. It explains how we construct the counterfactual average income in the absence of migration (w_{id}^o) and the counterfactual number of temples (N_{jd}^o) using the estimated coefficients from the reduced-form analysis. These quantities underpin the simulations discussed in Section 5 of the main text.

To construct the counterfactuals in Section 5, we estimate two key components: (i) the counterfactual average income in the absence of migration, denoted w_{id}^o , and (ii) the counterfactual number of Evangelical and Catholic temples, denoted N_{jd}^o , where $j = E$ for Evangelicals and $j = C$ for Catholics.

To compute w_{id}^o , we proceed in three steps. First, using the estimates from column (1) of Table 4, we calculate the counterfactual income growth rate for each municipality and period as

$$g_{dt}^0 = \Delta \ln(w_{dt}) - \hat{\beta}_t m_{dt},$$

where $\hat{\beta}_t$ is the estimated effect of migration on income growth for period t , $\ln(w_{dt})$ is the log of average income in municipality d and period t , and $\Delta \ln(w_{dt})$ is the observed change

in log income; i.e. the dependent variable in the equation that we used to estimate β . The term g_{dt}^0 captures how income would have evolved in the absence of migration.

Second, we use this growth rate to construct the counterfactual average income in the municipality as

$$w_{dt}^0 = w_{dt-1}(1 + g_{dt}^0),$$

where w_{dt-1} is the observed average income in the previous period.

Third, to obtain individual-level counterfactual income, we regress observed individual income on municipality dummies (separately for each period) and collect the residuals, denoted η_{idt} . We then define:²⁹

$$w_{id}^o = w_{dt}^0 + \eta_{idt}.$$

The construction of the counterfactual number of temples N_{jd}^o follows a similar logic. First, using the estimates from columns (2) and (4) of Table 5, we compute

$$h_{jdt}^0 = \Delta N_{jdt} - \hat{\gamma}_{jt} m_{dt},$$

where $\hat{\gamma}_{jt}$ is the estimated effect of migration on the number of temples of religious group j in period t , and ΔN_{jdt} is the observed variation in the number of temples. This gives us the change in temple count that would have occurred without migration. Then we compute the counterfactual number of temples as

$$N_{jdt}^o = N_{jdt-1} + h_{jdt}^0,$$

where N_{jdt-1} is the observed stock of temples in the previous period.

F More on the Sensitivity of the Demand to Income and Infrastructure

This Appendix presents additional evidence on the sensitivity of religious demand to changes in income and infrastructure. While Figure 7 offers useful insight, it is based on pre-migration counterfactuals, excludes the 1991 Census, and considers only one direction of variation—increasing income and reducing temple supply. Here, we adopt a more straightforward approach designed to infer the sensitivity of religious demand under different conditions.

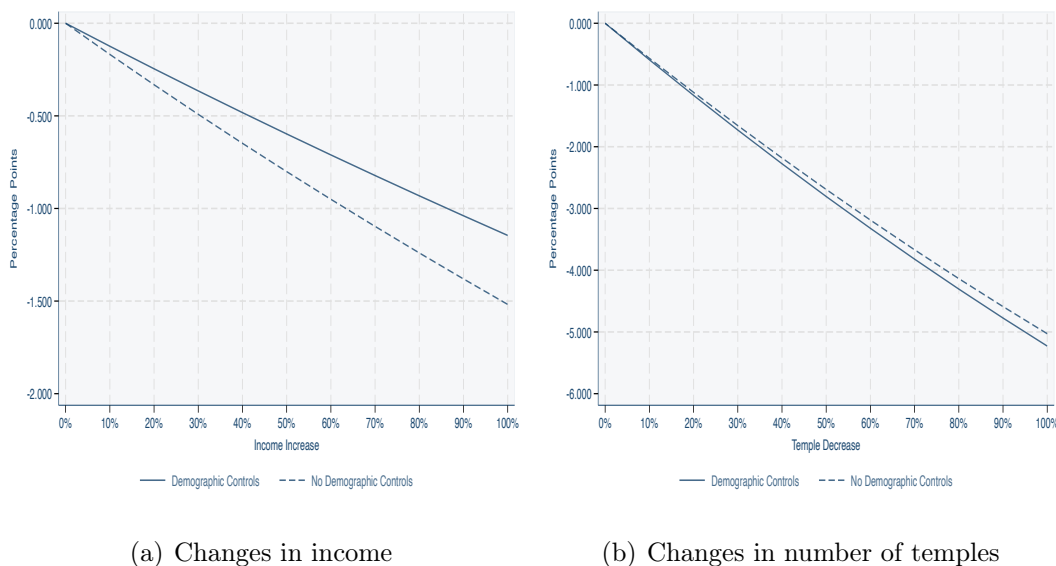
We begin by varying income. For each year (1991, 2000, 2010), we simulate proportional increases in individuals' observed income—unlike Figure 7, which considers only 2000 and 2010, uses pre-migration counterfactuals, and focuses on income declines. Specifically, we compute predicted changes in the Evangelical share as income increases by a factor $\varrho \in [1, 2]$, with $\varrho = 1$ denoting baseline income and $\varrho = 2$ a doubling. For each individual, we calculate the change in predicted Evangelical affiliation across income levels and average these differences using Census sampling weights. Figure 4(a) reports the results for models

²⁹To ensure comparability, we define observed income as $w_{id} = w_{dt} + \eta_{idt}$, using the same residual η_{idt} as in the counterfactual $w_{id}^o = w_{dt}^0 + \eta_{idt}$. This approach holds individual heterogeneity fixed and attributes the effect of migration solely to changes in the municipal income component.

estimated with and without demographic controls. In all cases, higher income is associated with lower Evangelical affiliation. For instance, doubling income reduces the Evangelical share by approximately 1.1 percentage points when demographic controls are included.

Next, we conduct a parallel simulation for religious infrastructure. Using the same framework, we simulate the effect of reducing the number of observed Evangelical temples by a factor $\varrho \in [0, 1]$, where $\varrho = 0$ corresponds to baseline temple supply and $\varrho = 1$ to complete removal. As in the income counterfactuals, we compute the predicted change in Evangelical affiliation for each individual and average across the sample using Census weights. Figure 4(b) reports the results. The decline in Evangelical affiliation is large and monotonic: eliminating half of all temples leads to a drop of about 2 percentage points in the Evangelical share, and full removal results in declines exceeding 4 percentage points. Notably, these effects are substantially larger than those observed in the income simulations. The observed patterns are similar across model specifications.

Figure F3: **Sensitivity of Evangelical Affiliation to Changes in Income and Infrastructure**



Panel (a) illustrates the sensitivity of the share of Evangelicals to proportional increases in income, with $\varrho \in [1, 2]$, where $\varrho = 1$ denotes baseline income and $\varrho = 2$ represents a doubling. Panel (b) shows the sensitivity to proportional decreases in the number of temples, with $\varrho \in [0, 1]$, where $\varrho = 0$ is the baseline and $\varrho = 1$ indicates complete removal of temples. In both panels, the y-axis reports the average change in Evangelical affiliation (in percentage points) *relative to the baseline scenario*—i.e., the predicted difference between each counterfactual and the observed data. The solid line corresponds to models estimated with demographic controls; the dashed line excludes them.